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Introducing metal–sulfur active sites in metal–organic frameworks via post-synthetic modification for hydrogenation catalysis

In the format provided by the authors and unedited

Table of Contents

Section A. Materials and General Methods

All reagents were obtained from commercial suppliers and were utilized as received without additional purification.

Nuclear Magnetic Resonance (NMR) spectroscopy was performed using a Bruker Avance III 500 spectrometer operating at a frequency of 500 MHz. Chemical shifts are reported in parts per million (ppm) relative to the residual solvent peaks of non-deuterated solvents: for 1H NMR, 7.26 ppm in $CDCl_3$ and 3.35, 4.78 ppm in CD_3OD.

Thermogravimetric Analysis (TGA) was conducted using a TGA/DCS 1 system from Mettler-Toledo AG, based in Schwerzenbach, Switzerland, which is operated via a PC using STARe software. Samples were heated at a rate of 5 °C per minute across specified temperature intervals under an air flow of 20 mL per minute.

UV-vis Absorption Spectra were obtained using a Shimadzu UV-3600 UV/vis/NIR spectrometer, which features an integrating sphere to compensate for scatter and reflectance effects during measurement.

Fourier-Transform Infrared Spectroscopy M easurements (FTIR) were conducted at the KECKII/NUANCE facility at Northwestern University using a Nicolet iS50 spectrometer by Thermo Nicolet. This spectrometer includes a tabletop optics module (TOM), which facilitates Polarization Modulation Infrared Reflection Absorption Spectroscopy (PM-IRRAS). Spectra were recorded over a wavelength range from 4000 to 400 cm^{-1}.

Scanning Electron Microscopy (SEM) and **Energy Dispersive Spectroscopy (EDS)** analyses were performed using a Hitachi SU8030 at the KECKII/NUANCE facility at Northwestern University. Samples were pre-coated with an 18 nm layer of osmium. The EDS data were analyzed using the New Oxford AZtec EDS software.

Section B. Synthetic Protocols and Characterization

Synthesis of powder samples of Zn-MFU-4*l*-Cl

The preparation of Zn-MFU-4*l*-Cl was performed according to the previously reported procedure.[1] In a 100-dram scintillation vial H_2BTDD (100 mg, 0.36 mmol), $ZnCl_2$ (1.0 g, 7.3 mmol) and DMF (60 mL) were heated for 18 h at 140 °C and cooled down to room temperature. The precipitate was collected by centrifuged and washed with 20 mL DMF for 3 times and followed by 20 mL MeOH for 3 times, and 20 mL CH2Cl2 for 3 times. Then the solid was dried for 24 h at 180 °C under vacuum. Yield brown microcrystalline powder.

Transmetalation of Zn-MFU-4*l*-Cl to Co-MFU-4*l*-Cl

The transmetallation of Zn-MFU-4*l*-Cl to Co-MFU-4*l* was performed according to the previously reported procedure.[1] Anhydrous cobalt (II) chloride (0.5 g, 2.1 mmol) was dissolved in DMF (16 mL) and MFU-4*l* (100 mg, 0.08 mmol) was added to the solution. The reaction mixture was stirred for 20 h at 140 °C in a 20-dram vail. The dark-green or dark-blue precipitate was centrifuged, washed with DMF, methanol and dichloromethane.

Transmetalation of Zn-MFU-4*l*-Cl to Ni-MFU-4*l*-Cl

The transmetallation of Zn-MFU-4*l*-Cl to Ni-MFU-4*l* was performed according to the previously reported procedure.[1] Nickel(II) chloride (0.6 g, 2.6 mmol) was dissolved in N,Ndimethylacetamide (16 mL) and MFU-4*l* (100 mg, 0.08 mmol) was added to the solution. The reaction mixture was heated for 20 h at 60 °C in a 100-dram scintillation vial. N,Ndimethylacetamide was exchanged with fresh N,N-dimethylacetamide and then added Nickel(II) chloride (0.4 g) and heated for 20 h at 60 °C. The green precipitate was centrifuged, washed with N,N-dimethylacetamide, methanol and dichloromethane.

Synthesis of powder samples of M-MFU-4*l*-OH

M-MFU-4*l*-Cl (50 mg) was treated with 50 mL of a 1 M aqueous solution of 4-ethylmorpholine for 12 hours. Subsequently, the samples were washed with water three times. The resulted powder sample was dried under N_2 flow and activated under 120 °C before adsorption measurements.

Post-synthetic conversion of the single-crystal samples of M-MFU-4*l*-Cl to M-MFU-4*l*-OH.

Single crystal samples of **M-MFU-4*l*-Cl** (approximately 2 mg) were treated with 4 mL of a 1 M aqueous solution of 4-ethylmorpholine for 12 hours. Subsequently, the samples were washed with water and MeOH three times. The morphology of the crystals remained unchanged, resulting in a pale green color for M = Ni and pale pink for M = Co, respectively.

Section C. Single-Crystal Structures

Supplementary Table 1. Crystal data and structure refinements of M_2X_2BBTA and M-MFU-4l-X where M = Co and Ni, X = Cl, OH and SH.

Identification code	Co_2Cl_2BBTA	$Co_2(OH)_2$BBTA	$Co_2(SH)_2$BBTA
Empirical formula	$C_6H_2Cl_2Co_2N_6O_2$	$C_6H_2Co_2N_6O_4$	$C_6H_2Co_2N_6S_7$
Formula weight	378.89	339.98	500.44
Temperature/K	100.15	100.15	100.15
Crystal system	trigonal	trigonal	trigonal
Space group	*R-3m*	*R-3m*	*R-3m*
a/Å	24.5743(9)	24.655(5)	23.891(2)
b/Å	24.5743(9)	24.655(5)	23.891(2)
c/Å	8.3838(2)	7.9848(18)	7.8380(9)
α/°	90	90	90
β/°	90	90	90
γ/°	120	120	120
Volume/Å^3	4384.6(3)	4204(2)	3874.3(8)
Z	9	9	9
ρ_{calc}g/cm^3	1.586	1.593	1.958
μ/mm^{-1}	16.202	14.436	23.065
F(000)	2088	2034	2250
Crystal size/mm^3	0.35 × 0.01 × 0.01	0.3 × 0.01 × 0.01	0.03 × 0.01 × 0.01
Radiation	CuKα (λ = 1.54184)	CuKα (λ = 1.54184)	CuKα (λ = 1.54184)
2Θ range for data collection/°	7.194 to 150.104	7.17 to 95.06	7.4 to 135.824
Index ranges	-25 ≤ h ≤ 29, -29 ≤ k ≤ 23, -10 ≤ l ≤ 7	-19 ≤ h ≤ 19, -20 ≤ k ≤ 23, -7 ≤ l ≤ 7	-28 ≤ h ≤ 28, -28 ≤ k ≤ 22, -9 ≤ l ≤ 6
Reflections collected	5782	2106	2839
Independent reflections	1075 [R_{int} = 0.0288, R_{sigma} = 0.0191]	477 [R_{int} = 0.1730, R_{sigma} = 0.1030]	850 [R_{int} = 0.1123, R_{sigma} = 0.0947]
Data/restraints/parameters	1075/0/45	477/0/46	850/21/60
Goodness-of-fit on F^2	1.159	1.045	1.053
Final R indexes [I>=2σ (I)]	R_1 = 0.0542, wR_2 = 0.1576	R_1 = 0.0886, wR_2 = 0.2431	R_1 = 0.0870, wR_2 = 0.2181
Final R indexes [all data]	R_1 = 0.0565, wR_2 = 0.1595	R_1 = 0.1050, wR_2 = 0.2546	R_1 = 0.1152, wR_2 = 0.2371
Largest diff. peak/hole / e Å^{-3}	1.44/-0.60	0.66/-0.62	1.37/-1.16
CCDC Number	2376631	2376626	2376634

* Empirical formula, formula weight and ρ_{calc} include the solvent molecules from the applied solvent masks.

Supplementary Table 1. Continued

Identification code	Ni_2Cl_2BBTA	$Ni_2(OH)_2BBTA$	$Ni_2(SH)_2BBTA$
Empirical formula	$C_6H_6Cl_2N_6Ni_2O_{3.66}$	$C_6H_6N_6Ni_2O_4$	$C_6H_{4.4}N_6Ni_2O_{2.38}S_2$
Formula weight	409.07	343.54	380.14
Temperature/K	293(2)	293(2)	293(2)
Crystal system	trigonal	trigonal	trigonal
Space group	*R-3m*	*R-3m*	*R-3m*
a/Å	24.456(14)	24.89(3)	24.35(4)
b/Å	24.456(14)	24.89(3)	24.35(4)
c/Å	8.099(3)	7.979(3)	8.098(4)
α/°	90	90	90
β/°	90	90	90
γ/°	120	120	120
Volume/Å^3	4195(5)	4281(10)	4159(13)
Z	9	9	9
ρ_{calc}g/cm^3	1.457	1.199	1.366
μ/mm^{-1}	0	0	0
F(000)	554	473	529
Crystal size/mm^3	0.0005 × 0.0002 × 0.0001	0.0004 × 0.0001 × 0.0001	0.0004 × 0.0001 × 0.00005
Radiation	electron diffractometer JEOL JSM-2300ED, LaB6 (λ = 0.0251)	electron diffractometer JEOL JSM-2300ED, LaB6 (λ = 0.0251)	electron diffractometer JEOL JSM-2300ED, LaB6 (λ = 0.0251)
2Θ range for data collection/°	0.118 to 1.438	0.116 to 1.348	0.118 to 1.3
Index ranges	$-20 \le h \le 21$, $-24 \le k \le 24$, $-8 \le l \le 8$	$-21 \le h \le 21$, $-23 \le k \le 23$, $-7 \le l \le 7$	$-18 \le h \le 19$, $-22 \le k \le 22$, $-7 \le l \le 7$
Reflections collected	3607	2648	2565
Independent reflections	541 [R_{int} = 0.4564, R_{sigma} = 0.2814]	460 [R_{int} = 0.2993, R_{sigma} = 0.2061]	397 [R_{int} = 0.3714, R_{sigma} = 0.2078]
Data/restraints/parameters	541/57/46	460/63/40	397/69/46
Goodness-of-fit on F^2	0.914	0.977	1.107
Final R indexes [I>=2σ (I)]	R_1 = 0.1164, wR_2 = 0.2580	R_1 = 0.1300, wR_2 = 0.3100	R_1 = 0.1547, wR_2 = 0.3496
Final R indexes [all data]	R_1 = 0.1690, wR_2 = 0.2990	R_1 = 0.1685, wR_2 = 0.3390	R_1 = 0.2078, wR_2 = 0.3842
Largest diff. peak/hole / e Å^{-3}	0.15/-0.20	0.32/-0.14	0.24/-0.37
CCDC Number	2385609	2385610	2385611

* Empirical formula, formula weight and ρ_{calc} include the solvent molecules from the applied solvent masks.

Supplementary Table 1. Continued

Identification code	Co-MFU-4*l*-Cl	Co-MFU-4*l*-OH	Co-MFU-4*l*-SH
Empirical formula	$C_{36}H_{12}Cl_4Co_4N_{18}O_{12.1}Zn$	$C_{64.8}H_{127.2}Co_4N_{18}O_{46.8}Zn$	$C_{36}H_{12}Co_4N_{18}O_{13.44}S_{3.44}Zn$
Formula weight	1380.33	2208.53	1322.93
Temperature/K	100.0(7)	100.0(10)	100.15
Crystal system	cubic	cubic	cubic
Space group	*Fm-3m*	*Fm-3m*	*Fm-3m*
a/Å	30.8165(5)	30.7820(10)	30.8219(14)
b/Å	30.8165(5)	30.7820(10)	30.8219(14)
c/Å	30.8165(5)	30.7820(10)	30.8219(14)
α/°	90	90	90
β/°	90	90	90
γ/°	90	90	90
Volume/Å^3	29265.1(14)	29167(3)	29280(4)
Z	8	8	8
ρ_{calc}g/cm^3	0.627	1.006	0.600
μ/mm^{-1}	4.587	4.255	4.356
F(000)	5443	9235	5236.0
Crystal size/mm^3	0.01 × 0.01 × 0.01	0.01 × 0.01 × 0.01	0.01 × 0.01 × 0.01
Radiation	Cu Kα (λ = 1.54184)	Cu Kα (λ = 1.54184)	CuKα (λ = 1.54184)
2Θ range for data collection/°	8.114 to 94.958	8.124 to 95.098	8.114 to 94.936
Index ranges	-28 ≤ h ≤ 28, -28 ≤ k ≤ 25, -29 ≤ l ≤ 29	-29 ≤ h ≤ 17, -26 ≤ k ≤ 29, -28 ≤ l ≤ 18	-20 ≤ h ≤ 25, -28 ≤ k ≤ 15, -29 ≤ l ≤ 10
Reflections collected	11442	9223	4123
Independent reflections	732 [R_{int} = 0.0816, R_{sigma} = 0.0295]	730 [R_{int} = 0.1234, R_{sigma} = 0.0489]	729 [R_{int} = 0.1519, R_{sigma} = 0.0978]
Data/restraints/parameters	732/54/65	730/52/54	729/77/64
Goodness-of-fit on F^2	1.098	1.062	0.868
Final R indexes [I>=2σ (I)]	R_1 = 0.0854, wR_2 = 0.2449	R_1 = 0.0493, wR_2 = 0.1394	R_1 = 0.0726, wR_2 = 0.1882
Final R indexes [all data]	R_1 = 0.1000, wR_2 = 0.2640	R_1 = 0.0741, wR_2 = 0.1622	R_1 = 0.1092, wR_2 = 0.2229
Largest diff. peak/hole / e Å^{-3}	0.98/-1.02	0.46/-0.28	0.43/-0.46
CCDC Number	2376630	2376629	2376628

* Empirical formula, formula weight and ρ_{calc} include the solvent molecules from the applied solvent masks.

Supplementary Table 1. Continued

Identification code	Ni-MFU-4*l*-Cl	Ni-MFU-4*l*-OH	Ni-MFU-4*l*-SH
Empirical formula	$C_{36}H_{122.4}Cl_4N_{18}Ni_{2.5}O_{69.08}Zn_{2.5}$	$C_{36}H_{79.2}N_{18}Ni_{2.5}O_{49.54}Zn_{2.5}$	$C_{55.2}H_{88.8}N_{18}Ni_{2.5}O_{32.86}S_3Zn_{2.5}$
Formula weight	2365.13	1867.18	1936.71
Temperature/K	100.15	100.15	100
Crystal system	cubic	cubic	cubic
Space group	*Fm-3m*	*Fm-3m*	*Fm-3m*
a/Å	30.7457(7)	30.7537(18)	30.8207(5)
b/Å	30.7457(7)	30.7537(18)	30.8207(5)
c/Å	30.7457(7)	30.7537(18)	30.8207(5)
α/°	90	90	90
β/°	90	90	90
γ/°	90	90	90
Volume/Å^3	29064(2)	29087(5)	29277.1(15)
Z	8	8	8
ρ_{calc}g/cm^3	1.081	0.853	0.879
μ/mm^{-1}	2.202	1.35	1.657
F(000)	9840	7700	8015
Crystal size/mm^3	0.01 × 0.01 × 0.01	0.01 × 0.01 × 0.01	0.01 × 0.01 × 0.01
Radiation	CuKα (λ = 1.54184)	CuKα (λ = 1.54184)	CuKα (λ = 1.54184)
2Θ range for data collection/°	8.134 to 95.246	8.132 to 95.214	8.114 to 95.23
Index ranges	$-28 \le h \le 28$, $-17 \le k \le 28$, $-23 \le l \le 29$	$-25 \le h \le 27$, $-17 \le k \le 29$, $-28 \le l \le 25$	$-18 \le h \le 28$, $-29 \le k \le 29$, $-27 \le l \le 17$
Reflections collected	7229	6499	7744
Independent reflections	728 [R_{int} = 0.0939, R_{sigma} = 0.0421]	729 [R_{int} = 0.1918, R_{sigma} = 0.0925]	732 [R_{int} = 0.0549, R_{sigma} = 0.0260]
Data/restraints/parameters	728/29/68	729/89/61	732/34/68
Goodness-of-fit on F^2	1.035	0.911	1.088
Final R indexes [I>=2σ (I)]	R_1 = 0.0482, wR_2 = 0.1469	R_1 = 0.0633, wR_2 = 0.1759	R_1 = 0.0498, wR_2 = 0.1657
Final R indexes [all data]	R_1 = 0.0622, wR_2 = 0.1542	R_1 = 0.0942, wR_2 = 0.1928	R_1 = 0.0584, wR_2 = 0.1731
Largest diff. peak/hole / e Å^{-3}	0.24/-0.33	0.57/-0.32	0.44/-0.31
CCDC Number	2376627	2376633	2376632

* Empirical formula, formula weight and ρ_{calc} include the solvent molecules from the applied solvent masks.

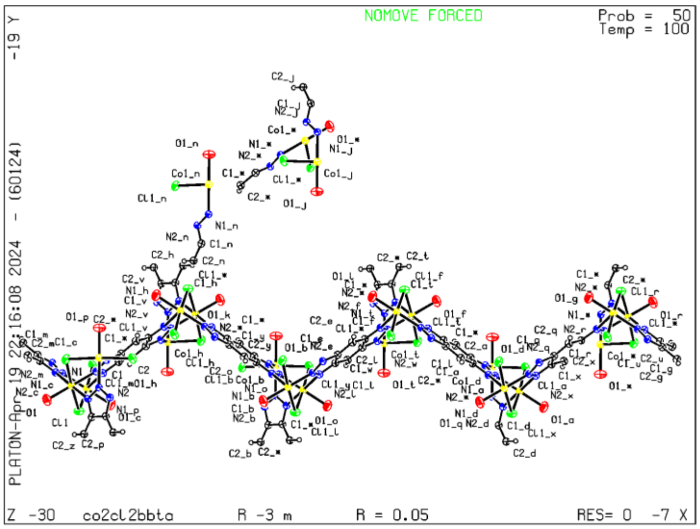
ORTEP-style illustrations and A- and/or B-level Alerts Justifications

Co_2Cl_2BBTA:

CCDC Number: 2376631

A- and/or B-level Alerts: None

ORTEP-style illustrations:



$Co_2(OH)_2$BBTA:

CCDC Number: 2376626

A- and/or B-level Alerts:

Problem: THETM01_ALERT_3_A The value of sine(theta_max)/wavelength is less than 0.550

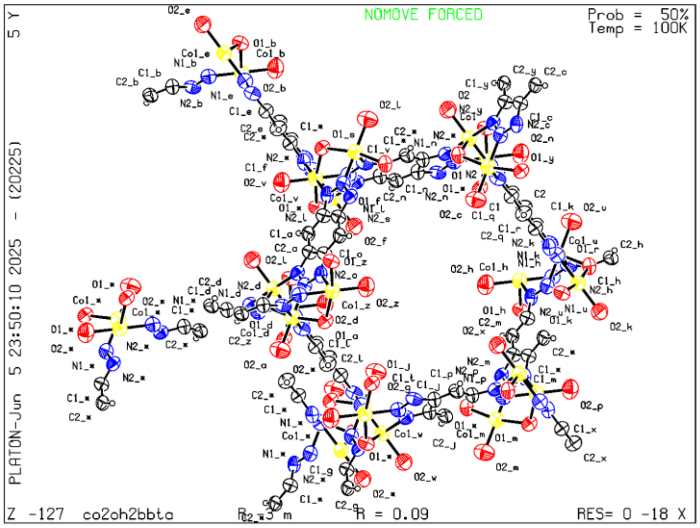
Calculated sin(theta_max)/wavelength = 0.4784

Response: The crystal was a very small needle, which resulted in weak diffraction. As a consequence, data could only be collected up to a resolution of 1.05 Å.

Problem: PLAT341_ALERT_3_B Low Bond Precision on C-C Bonds .... 0.0175 Ang.

Response: The reduced bond precision is attributed to the limited resolution of the dataset, which stems from the poor diffraction quality of the crystal.

ORTEP-style illustrations:

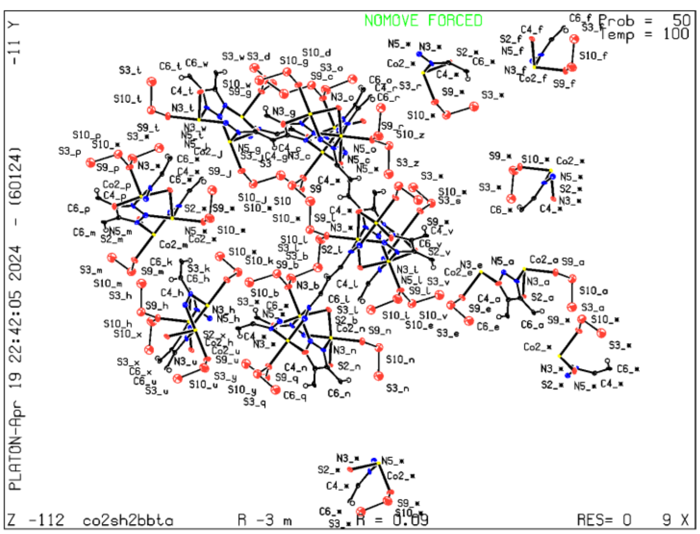


$Co_2(SH)_2$BBTA:

CCDC Number: 2376634

A- and/or B-level Alerts: None

ORTEP-style illustrations:



Ni_2Cl_2BBTA:

CCDC Number: 2385609

A- and/or B-level Alerts:

Problem: RINTA01_ALERT_3_A The value of Rint is greater than 0.25 Rint given 0.456

Response: The dataset was obtained by electron diffraction. The high Rint value arises from weak and limited-resolution diffraction, which is inherent to the small crystal size.

Problem: THETM01_ALERT_3_A The value of sine(theta_max)/wavelength is less than 0.550

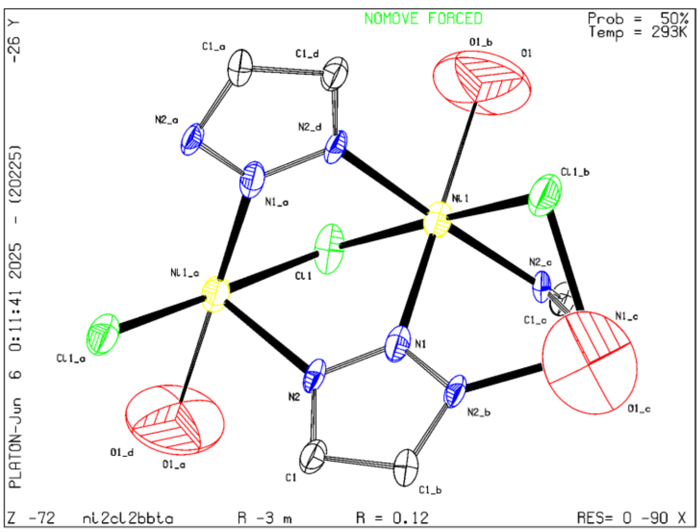
Calculated sin(theta_max)/wavelength = 0.4999

Response: Due to the small crystal size, diffraction was weak, limiting the resolution of the data. This measurement was acquired using electron diffraction, which often results in a lower sin(theta_max)/λ value compared to X-ray diffraction.

Problem: PLAT020_ALERT_3_A The Value of Rint is Greater Than 0.12 .... 0.456 Report

Response: As noted above, the elevated Rint is a result of weak diffraction and limited resolution associated with electron diffraction of a very small crystal.

ORTEP-style illustrations:



$Ni_2(OH)_2BBTA$:

CCDC Number: 2385610

A- and/or B-level Alerts:

Problem: RINTA01_ALERT_3_A The value of Rint is greater than 0.25 Rint given 0.299

Response: The dataset was collected using electron diffraction. The elevated Rint value is due to weak diffraction intensity and limited resolution, both resulting from the small crystal size.

Problem: THETM01_ALERT_3_A The value of sine(theta_max)/wavelength is less than 0.550

Calculated sin(theta_max)/wavelength = 0.4687

Response: The relatively low sin(theta _max)/λ is attributed to the weak diffraction and limited resolution inherent to data collected from a small crystal via electron diffraction.

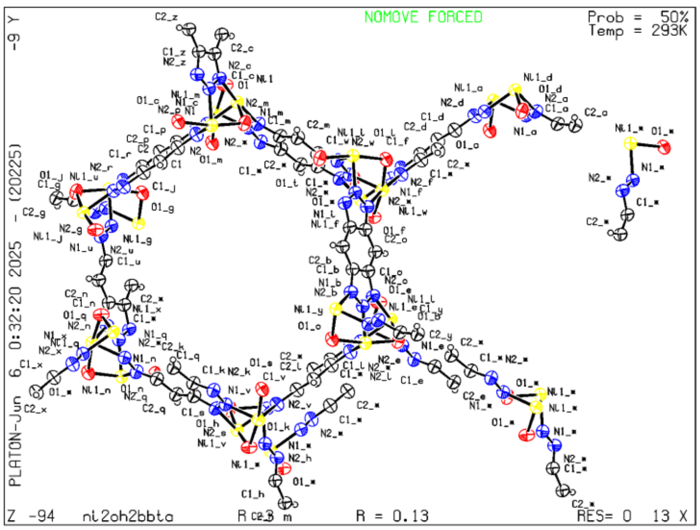
Problem: PLAT020_ALERT_3_A The Value of Rint is Greater Than 0.12 .... 0.299 Report

Response: As noted, the high Rint value results from weak diffraction and limited resolution due to the small crystal size and the use of electron diffraction.

Problem: PLAT341_ALERT_3_B Low Bond Precision on C-C Bonds .... 0.018 Ang.

Response: The limited precision of C–C bond lengths is due to the weak diffraction and restricted resolution, which are characteristic of electron diffraction data obtained from very small crystals.

ORTEP-style illustrations:



$Ni_2(SH)_2BBTA$:

CCDC Number: 2385611

A- and/or B-level Alerts:

Problem: RINTA01_ALERT_3_A The value of Rint is greater than 0.25

Rint given 0.371

Response: The data were collected by electron diffraction. The relatively high Rint value is due to weak diffraction and limited resolution caused by the small crystal size.

Problem: THETM01_ALERT_3_A The value of sine(theta_max)/wavelength is less than 0.550

Calculated sin(theta_max)/wavelength = 0.4520

Response: The dataset was collected by electron diffraction. The relatively low sin(theta_max)/wavelength is due to weak diffraction and limited resolution from the small crystal.

Problem: PLAT020_ALERT_3_A The Value of Rint is Greater Than 0.12 .... 0.371 Report

Response: As noted above, the high Rint value results from weak diffraction and limited resolution due to the small crystal and use of electron diffraction.

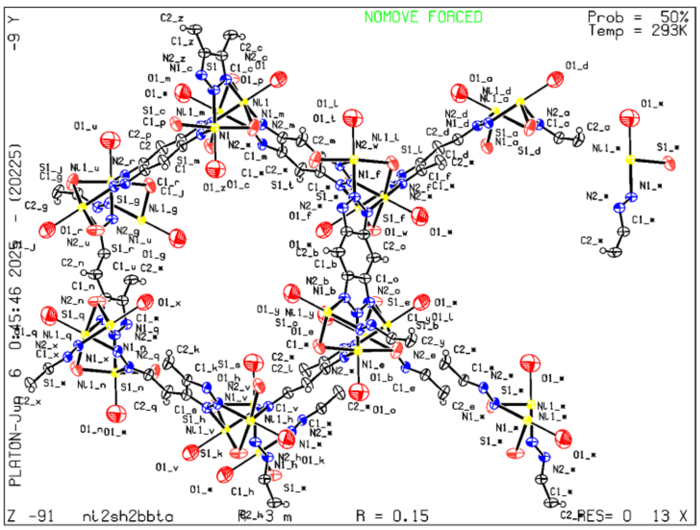
Problem: High wR2 Value (i.e. > 0.25) .... 0.38 Report

Response: The high wR2 value is attributed to weak diffraction and limited resolution, which are common for small crystals measured by electron diffraction.

Problem: PLAT341_ALERT_3_B Low Bond Precision on C-C Bonds .... 0.03 Ang.

Response: The low bond precision is due to weak diffraction and limited resolution, both of which result from the small crystal and use of electron diffraction.

ORTEP-style illustrations:



Co-MFU-4*l*-Cl:

CCDC Number: 2376630

A- and/or B-level Alerts:

Problem: THETM01_ALERT_3_A The value of sine(theta_max)/wavelength is less than 0.550

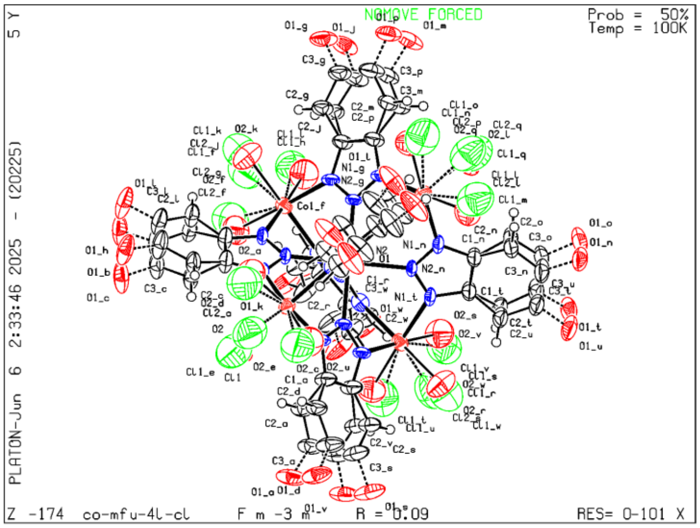
Calculated sin(theta_max)/wavelength = 0.4780

Response: The diffraction data were collected from a very small single crystal, which resulted in weak high-angle reflections. As a result, data could only be collected up to 1.05 Å resolution, leading to a lower sin(theta_max)/wavelength value.

Problem: PLAT242_ALERT_2_B Low 'MainMol' Ueq as Compared to Neighbors of Co1 Check

Response: The lower Ueq for Co1 is attributed to structural disorder between 4-coordinate and 6-coordinate Co sites. This local disorder affects the refinement of atomic displacement parameters and results in Ueq values that deviate from neighboring atoms.

ORTEP-style illustrations:



Co-MFU-4*l*-OH:

CCDC Number: 2376629

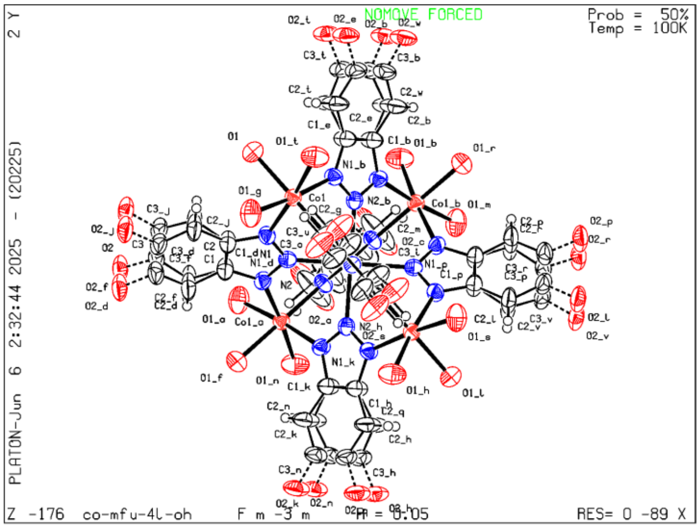
A- and/or B-level Alerts:

Problem: THETM01_ALERT_3_A The value of sine(theta_max)/wavelength is less than 0.550

Calculated sin(theta_max)/wavelength = 0.4786

Response: The diffraction data were collected from a very small single crystal, which resulted in weak high-angle reflections. As a result, data could only be collected up to 1.05 Å resolution, leading to a lower sin(theta_max)/wavelength value.

ORTEP-style illustrations:



Co-MFU-4*l*-SH:

CCDC Number: 2376628

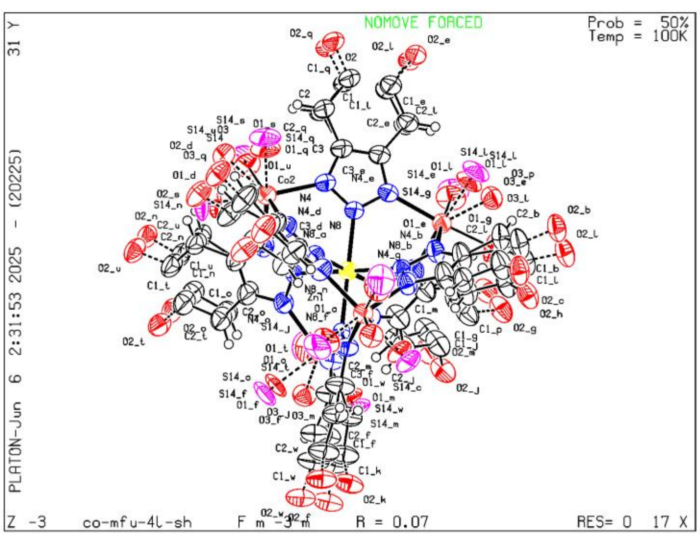
A- and/or B-level Alerts:

Problem: THETM01_ALERT_3_A The value of sine(theta_max)/wavelength is less than 0.550

Calculated sin(theta_max)/wavelength = 0.4779

Response: The diffraction data were collected from a very small single crystal, which resulted in weak high-angle reflections. As a result, data could only be collected up to 1.05 Å resolution, leading to a lower sin(theta_max)/wavelength value.

ORTEP-style illustrations:



Ni-MFU-4*l*-Cl:

CCDC Number: 2376627

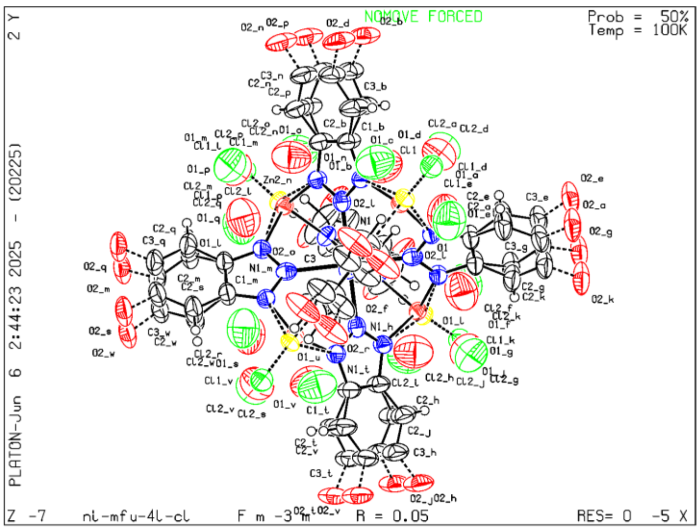
A- and/or B-level Alerts:

Problem: THETM01_ALERT_3_A The value of sine(theta_max)/wavelength is less than 0.550

Calculated sin(theta_max)/wavelength = 0.4791

Response: The diffraction data were collected from a very small single crystal, which resulted in weak high-angle reflections. As a result, data could only be collected up to 1.05 Å resolution, leading to a lower sin(theta_max)/wavelength value.

ORTEP-style illustrations:



Ni-MFU-4*l*-OH:

CCDC Number: 2376633

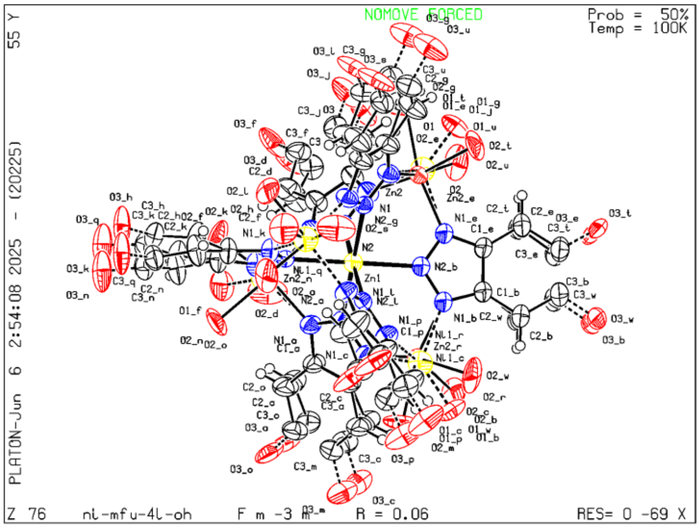
A- and/or B-level Alerts:

Problem: THETM01_ALERT_3_A The value of sine(theta_max)/wavelength is less than 0.550

Calculated sin(theta_max)/wavelength = 0.4790

Response: The diffraction data were collected from a very small single crystal, which resulted in weak high-angle reflections. As a result, data could only be collected up to 1.05 Å resolution, leading to a lower sin(theta_max)/wavelength value.

ORTEP-style illustrations:



Problem: RINTA01_ALERT_3_B The value of Rint is greater than 0.18. Rint given 0.192

Response: The relatively high Rint value is due to weak diffraction and limited resolution caused by the small crystal size.

Ni-MFU-4*l*-SH:

CCDC Number: 2376632

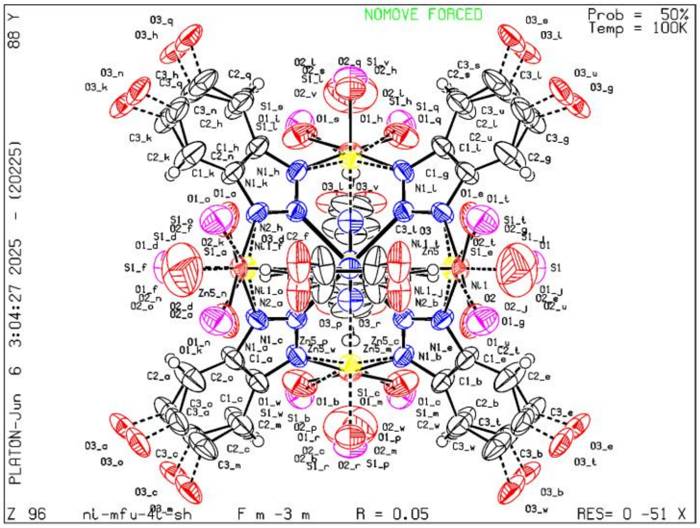
A- and/or B-level Alerts:

Problem: THETM01_ALERT_3_A The value of sine(theta_max)/wavelength is less than 0.550

Calculated sin(theta_max)/wavelength = 0.4791

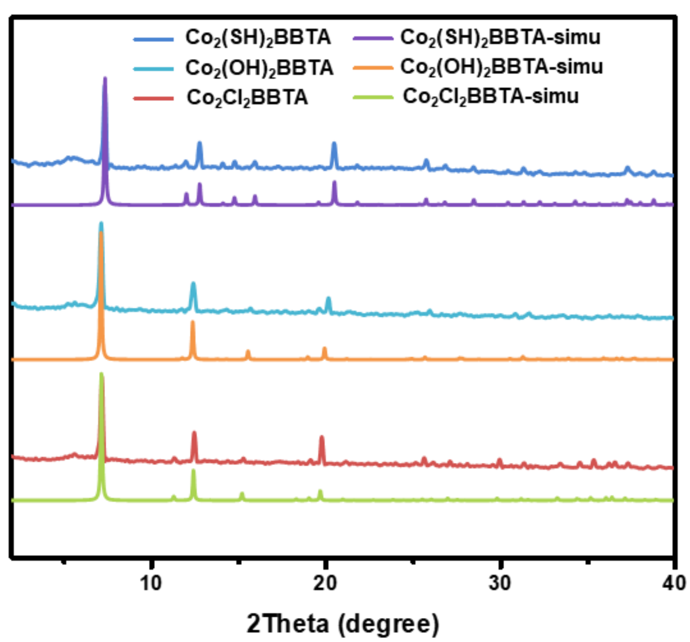
Response: The diffraction data were collected from a very small single crystal, which resulted in weak high-angle reflections. As a result, data could only be collected up to 1.05 Å resolution, leading to a lower sin(theta_max)/wavelength value.

ORTEP-style illustrations:

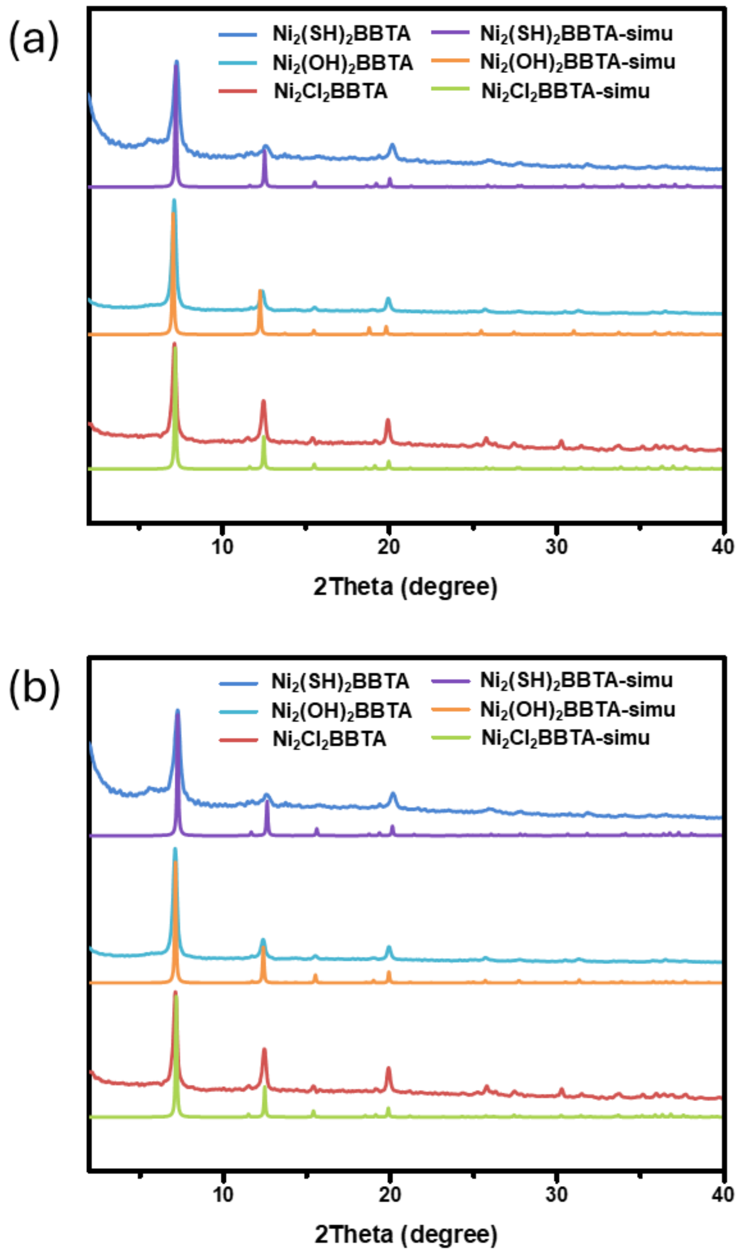


Supplementary Table 2. Refined Unit cell parameters of Ni_2X_2BBTA where X = Cl, OH and SH from PXRD.

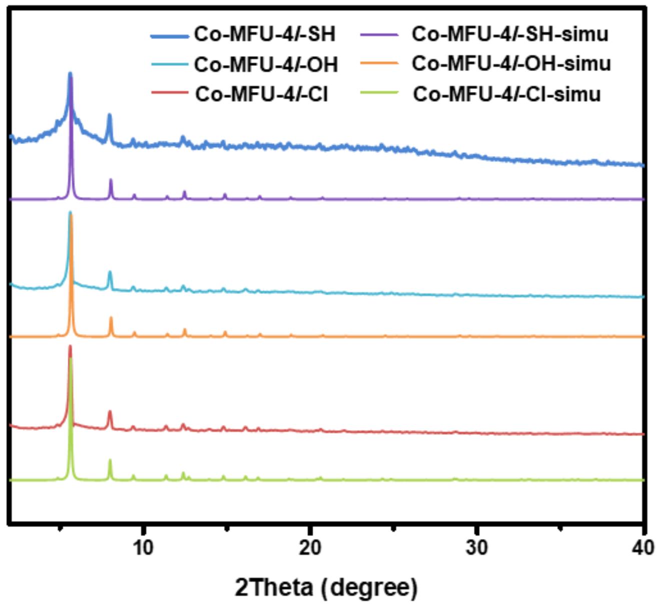
	Ni_2Cl_2BBTA	$Ni_2(OH)_2BBTA$	$Ni_2(SH)_2BBTA$
Space group	R-3m	R-3m	R-3m
a/Å	24.441(6)	24.608(4)	24.16(2)
b/Å	24.441(6)	24.608(4)	24.16(2)
c/Å	8.200(3)	8.021(2)	8.087(7)
α/°	90	90	90
β/°	90	90	90
γ/°	120	120	120
Volume/Å^3	4242(4)	4206(2)	4088(10)



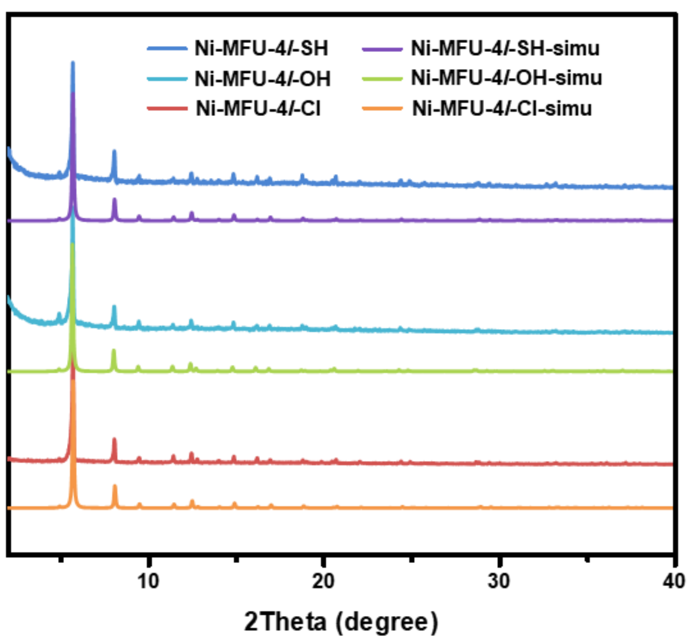
Supplementary Figure 1. Experimental and simulated PXRD of Co_2X_2BBTA, where X = Cl, OH and SH.



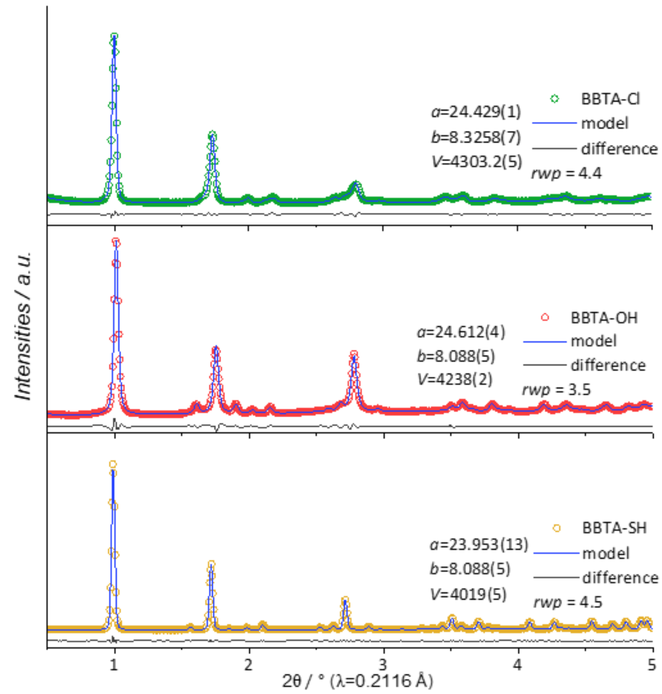
Supplementary Figure 2. Experimental and simulated PXRD patterns (a) from electron diffractions and (b) unit cell refinements of Ni_2X_2BBTA, where X = Cl, OH and SH.



Supplementary Figure 3. Experimental and simulated PXRD of Co-MFU-4l-X, where X = Cl, OH and SH.



Supplementary Figure 4. Experimental and simulated PXRD of Ni-MFU-4*l*-X, where X = Cl, OH and SH.



Supplementary Figure 5. Refinement of trigonal R -3 m model against PXRD data of Co_2X_2BBTA, where X = Cl, OH and SH, after individual ligand substitution procedures.

Section D. Sorption Properties

The theoretical pore volume (PV_{calc}) of the MOFs were calculated by the equation shown below:

$$PV_{calc} = \frac{n}{\rho}$$

Where n is the void rate that calculated by Mercury software with probe size as 1.8 Å and grid spacing as 0.3 Å; ρ is the theoretical density of the MOFs calculated from cif files. The theoretical N_2 sorption amounts, A_{calc}, were calculated with the equation as below:

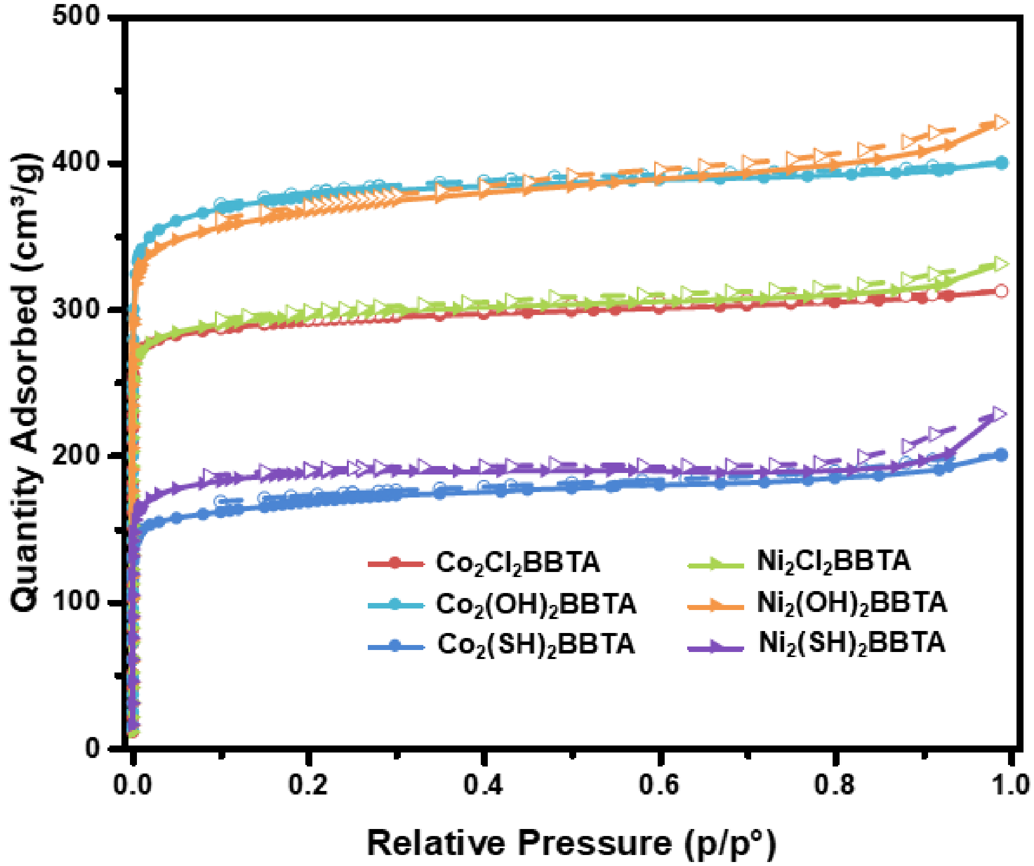
$$A_{calc} = \frac{PV_{calc} \cdot \rho_{N_2,l}}{M_{N_2}} \cdot V_{molar}$$

Where $\rho_{N_2,l}$ is the density of liquid N_2 at 77 K, M_{N_2} is the molar mass of N_2 and V_{molar} is the molar volume of a gas.

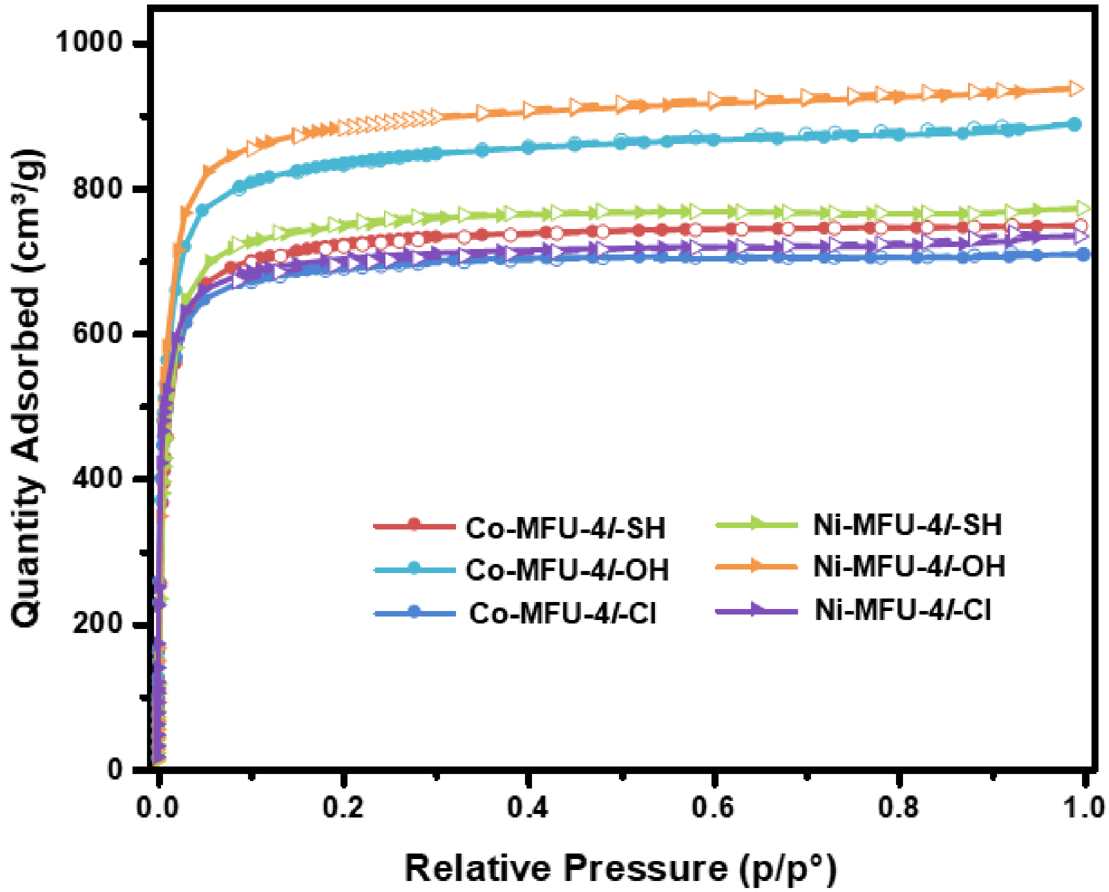
Supplementary Table 3. Theoretical and experimental porosity data of M_2X_2BBTA and M-MFU-4*l*-X where M = Co and Ni, X = Cl, OH and SH.

MOF	*n*(%)	*ρ* (g cm^{-3})	PV_{calc}(cm^3 g^{-1})	PV_{exp} (cm^3 g^{-1})	A_{calc} (cm^3 g^{-1})	A_{exp} (cm^3 g^{-1})	*BET Area* (m^2 g^{-1})	*Langmuir surface area* (m^2 g^{-1})
Co_2Cl_2BBTA	58.1	1.182	0.491	0.482	317	312	1180	1195
$Co_2(OH)_2$BBTA	58.9	1.095	0.538	0.612	347	400	1490	1470
$Co_2(SH)_2$BBTA	54.6	1.312	0.416	0.294	269	200	650	640
$Co_2(SH)_2$BBTA[a]	38.8	1.435	0.270	0.294	175	200	650	640
Ni_2Cl_2BBTA	57.7	1.234	0.468	0.483	302	330	1190	1195
$Ni_2(OH)_2$BBTA	60.9	1.047	0.582	0.658	376	427	1420	1470
$Ni_2(SH)_2$BBTA	57.3	1.221	0.469	0.297	303	228	730	705
$Ni_2(SH)_2$BBTA[a]	47.7	1.228	0.388	0.297	250	228	730	705
Co-MFU-4*l*-Cl	71.7	0.627	1.144	1.154	738	746	2750	-
Co-MFU-4*l*-OH	74.8	0.564	1.327	1.411	857	910	3480	-
Co-MFU-4*l*-SH	72.0	0.608	1.184	1.094	764	702	2660	-
Ni-MFU-4*l*-Cl	71.7	0.601	1.193	1.190	770	768	2900	-
Ni-MFU-4*l*-OH	74.6	0.550	1.356	1.449	875	940	3440	-
Ni-MFU-4*l*-SH	72.7	0.605	1.202	1.141	776	726	2720	-

a with a H_2O terminal group

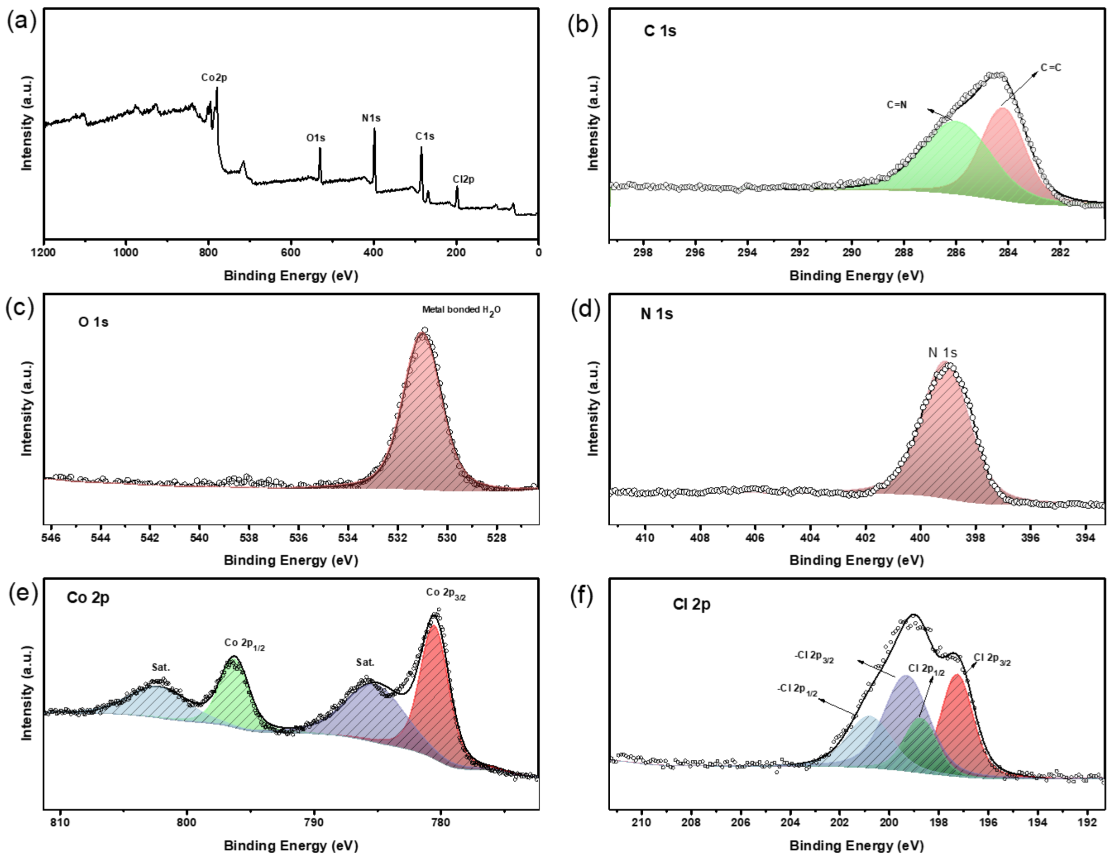


Supplementary Figure 6. N_2 isotherms of M_2X_2BBTA at 77 K (where M = Co and Ni, X = Cl, OH and SH)



Supplementary Figure 7. N_2 isotherms of M-MFU-4*l*-X at 77 K (where M = Co and Ni, X = Cl, OH and SH)

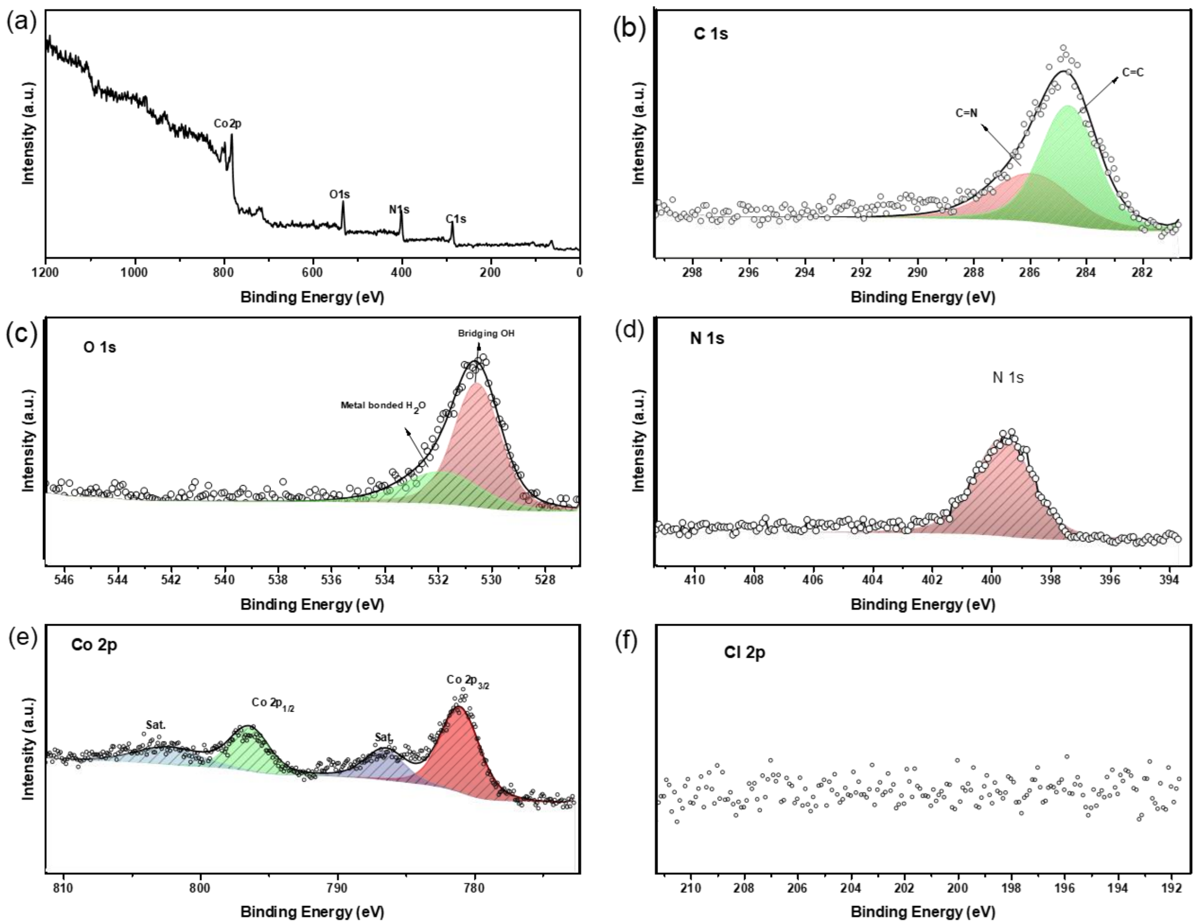
Section E. X-ray photoelectron spectroscopy (XPS)



Supplementary Figure 8. XPS spectra of Co_2Cl_2BBTA: (a) survey, (b) C1s, (c)O1s, (d)N1s, (e) Co2p and (f) Cl2p.

Supplementary Table 4. Summary of XPS spectra peaks of Co_2Cl_2BBTA

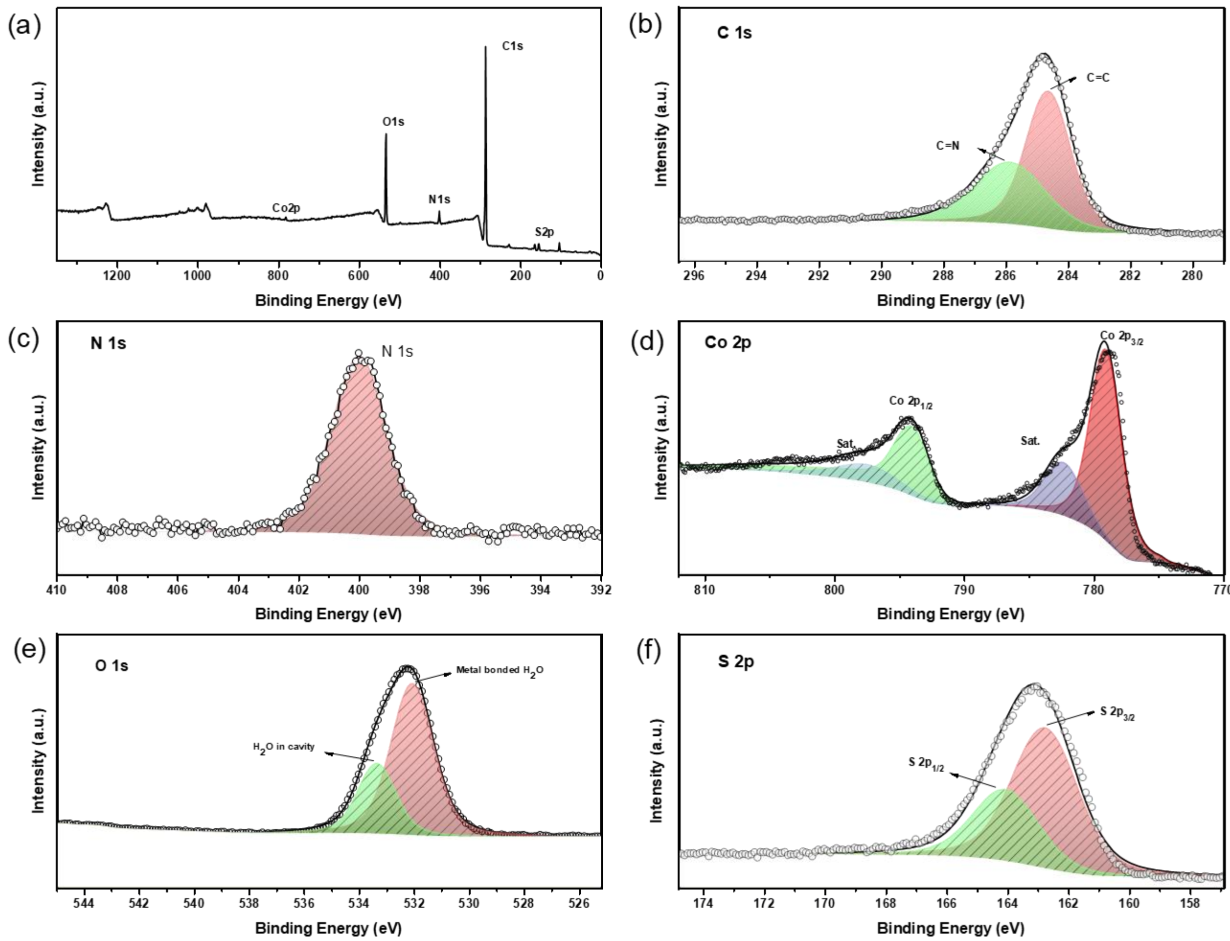
Element	Chemical State	Bonding Energy(eV)	Atomic % per Co
	C=C	284.15	
C	C=N	285.93	
O	H_2O (coordinates to Co)	530.99	
N	N=C/N=N	399.98	
	Cl^- $2p_{3/2}$	197.22	
	Cl^- $2p_{1/2}$	198.72	
	-Cl $2p_{3/2}$	199.26	
Cl	-Cl $2p_{1/2}$	200.76	91%
	Co^{2+} $2p_{3/2}$	780.46	
	Co^{2+} $2p_{1/2}$	796.25	
	Co^{2+} $2p_{3/2, sat.}$	785.15	
Co	Co^{2+} $2p_{1/2, sat.}$	802.27	100%



Supplementary Figure 9. XPS spectra of $Co_2(OH)_2$BBTA: (a) survey, (b) C1s, (c)O1s, (d)N1s, (e) Co2p and (f) Cl2p.

Supplementary Table 5. Summary of XPS spectra peaks of $Co_2(OH)_2$BBTA

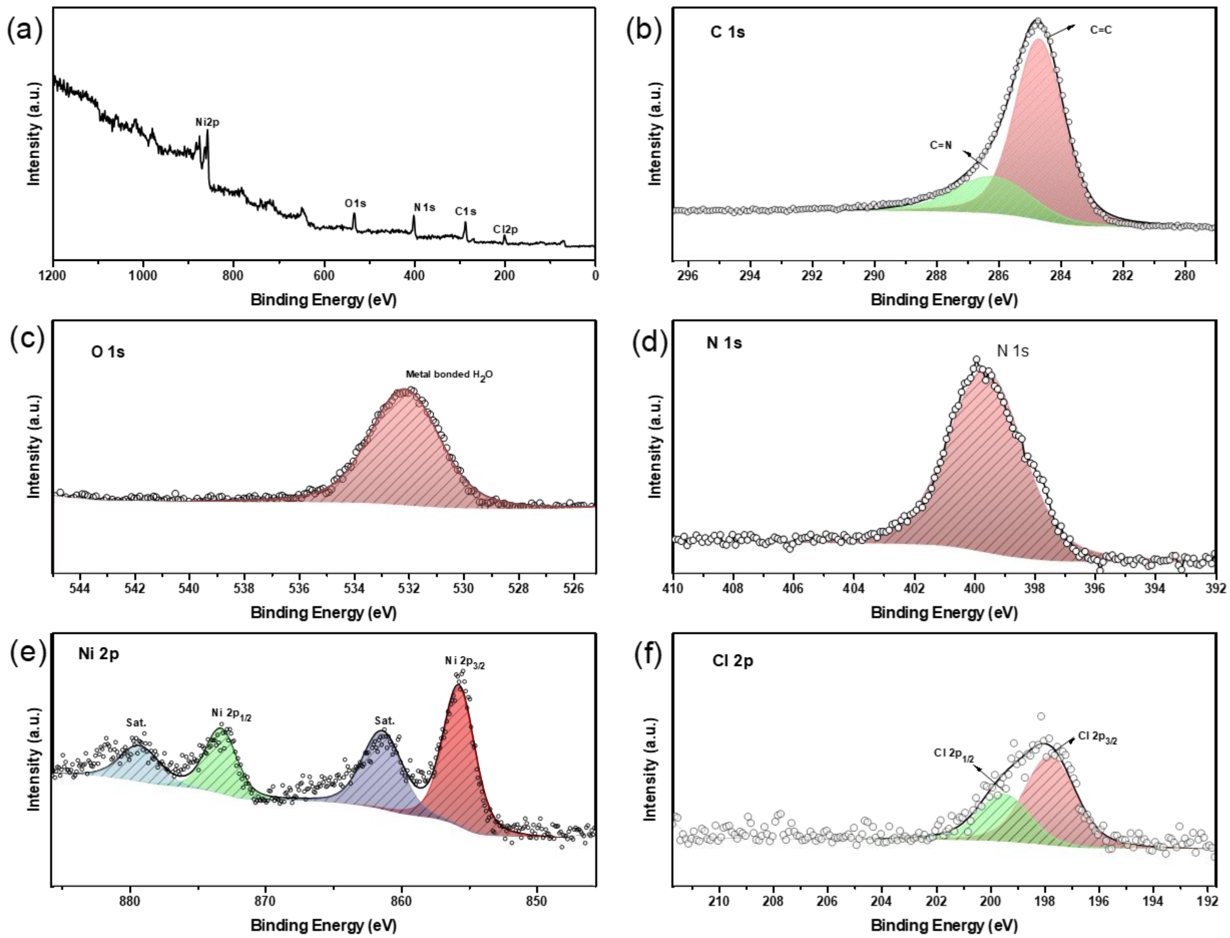
Element	Chemical State	Bonding Energy(eV)	Atomic % per Co
	C=C	285.91	
C	C=N	284.60	
	μ_2-OH	530.53	
O	H_2O (coordinates to Co)	531.84	
N	N=C/N=N	399.58	
Cl	N/A	N/A	0%
	Co^{2+} $2p_{3/2}$	781.02	
	Co^{2+} $2p_{1/2}$	796.40	
	Co^{2+} $2p_{3/2, sat.}$	786.49	
Co	Co^{2+} $2p_{1/2, sat.}$	802.28	100%



Supplementary Figure 10. XPS spectra of $Co_2(SH)_2$BBTA: (a) survey, (b) C1s, (c)N1s, (d) Co2p, (e) O1s and (f) S2p.

Supplementary Table 6. Summary of XPS spectra peaks of $Co_2(SH)_2$BBTA

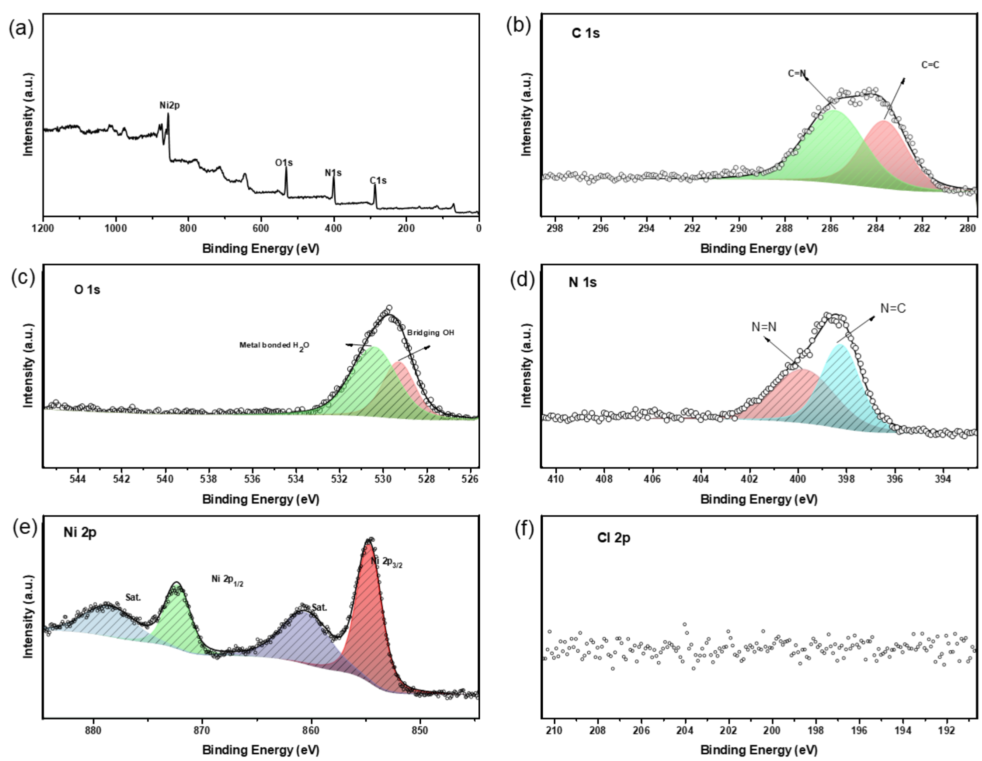
Element	Chemical State	Bonding Energy(eV)	Atomic % per Co
	C=C	284.61	
C	C=N	285.84	
	H_2O (coordinates to Co)	532.05	
O	H_2O (in cavity)	533.35	
N	N=C/N=N	399.98	
	S^{2-} $2p_{3/2}$	162.74	
S	S^{2-} $2p_{1/2}$	164.04	92%
	Co^{2+} $2p_{3/2}$	779.01	
	Co^{2+} $2p_{1/2}$	793.99	
	Co^{2+} $2p_{3/2, sat.}$	782.29	
Co	Co^{2+} $2p_{1/2, sat.}$	797.35	100%



Supplementary Figure 11. XPS spectra of Ni_2Cl_2BBTA: (a) survey, (b) C1s, (c)O1s, (d)N1s, (e) Ni2p and (f) Cl2p.

Supplementary Table 7. Summary of XPS spectra peaks of Ni_2Cl_2BBTA

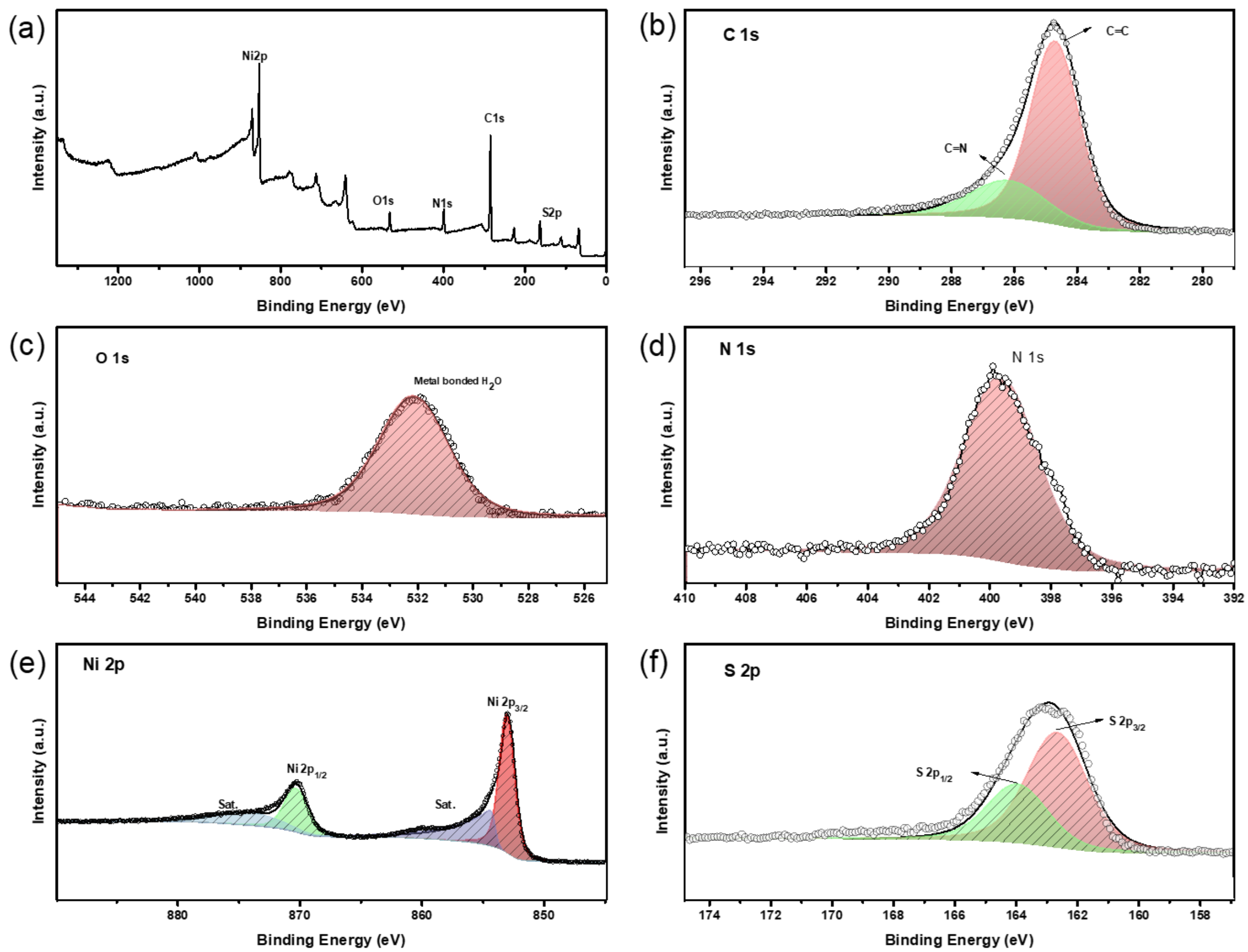
Element	Chemical State	Bonding Energy(eV)	Atomic % per Ni
	C=C	284.67	
C	C=N	287.48	
O	H_2O (in cavity)	531.29	
N	N=C/N=N	399.58	
	Cl^- $2p_{3/2}$	197.77	
Cl	Cl^- $2p_{1/2}$	199.47	98%
	Ni^{2+} $2p_{3/2}$	855.70	
	Ni^{2+} $2p_{1/2}$	873.24	
	Ni^{2+} $2p_{3/2, sat.}$	861.37	
Ni	Ni^{2+} $2p_{1/2, sat.}$	879.26	100%



Supplementary Figure 12. XPS spectra of $Ni_2(OH)_2BBTA$: (a) survey, (b) C1s, (c)O1s, (d)N1s, (e) Ni2p and (f) Cl2p.

Supplementary Table 8. Summary of XPS spectra peaks of $Ni_2(OH)_2BBTA$

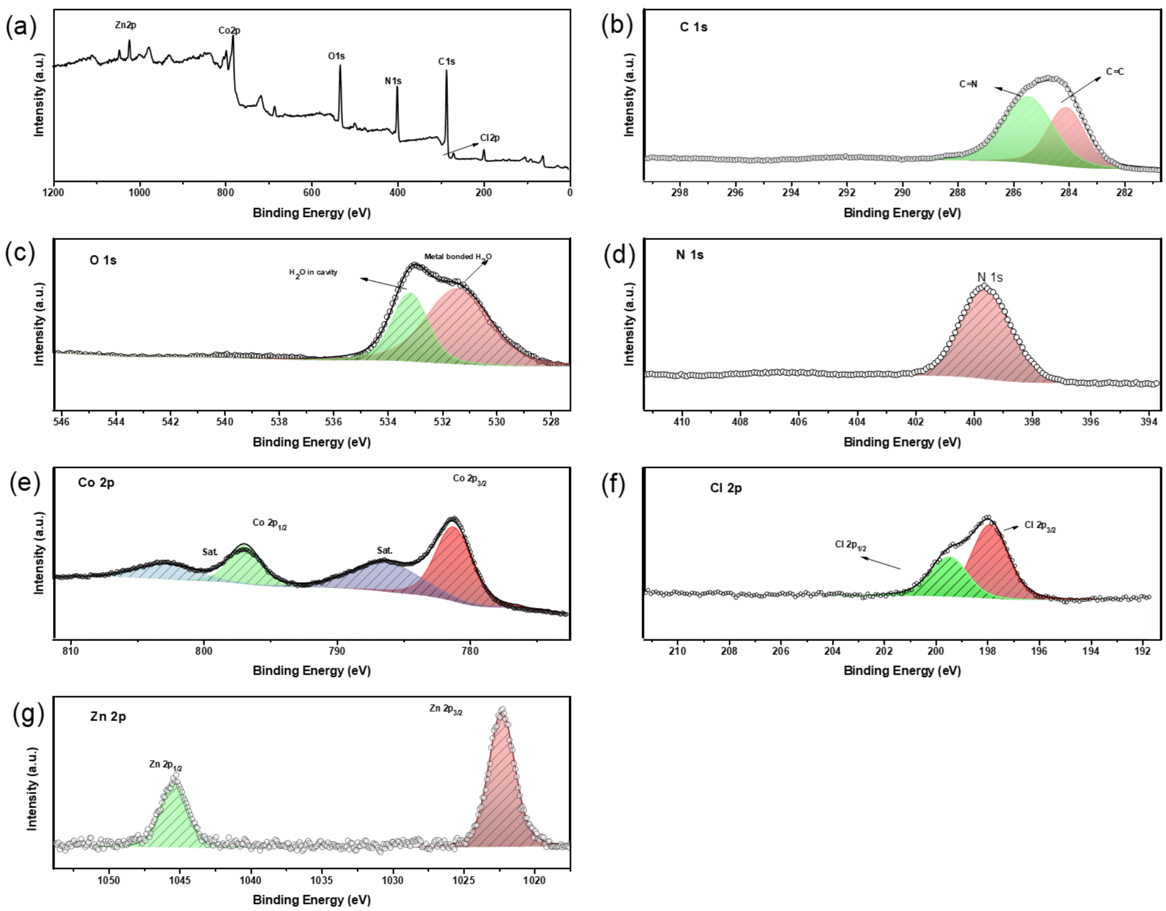
Element	Chemical State	Bonding Energy(eV)	Atomic % per Ni
	C=C	283.63	
C	C=N	285.79	
	μ_2-OH	529.27	
O	H_2O (coordinates to Co)	530.4	
	N=C	399.74	
N	N=N	398.22	
Cl	N/A	N/A	0%
	Ni^{2+} $2p_{3/2}$	854.65	
	Ni^{2+} $2p_{1/2}$	872.24	
	Ni^{2+} $2p_{3/2,\ sat.}$	860.52	
Ni	Ni^{2+} $2p_{1/2,\ sat.}$	878.52	100%



Supplementary Figure 13. XPS spectra of $Ni_2(SH)_2BBTA$: (a) survey, (b) C1s, (c)O1s, (d)N1s, (e) Ni2p and (f) S2p.

Supplementary Table 9. Summary of XPS spectra peaks of $Ni_2(SH)_2BBTA$

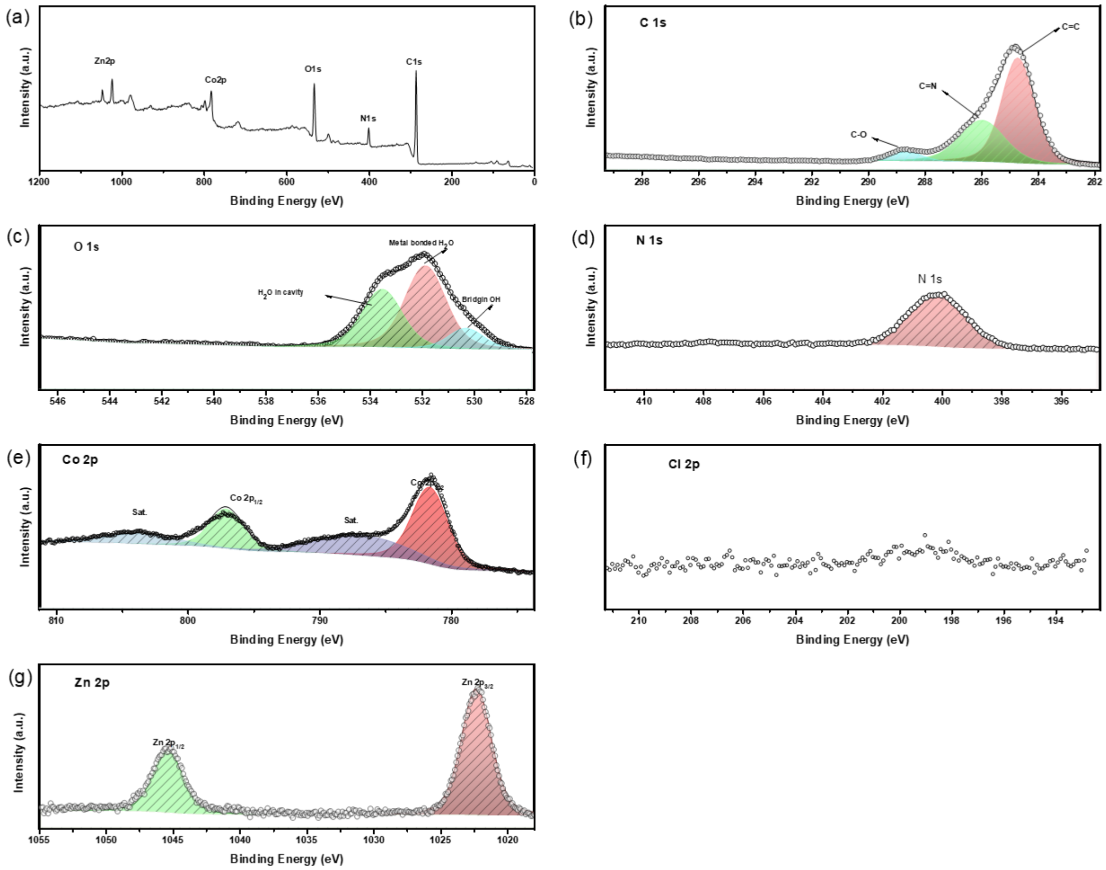
Element	Chemical State	Bonding Energy(eV)	Atomic % per Ni
	C=C	284.66	
C	C=N	286.17	
O	H_2O (in cavity)	532.13	
N	N=C/N=N	399.59	
	S^{2-} $2p_{3/2}$	162.61	
S	S^{2-} $2p_{1/2}$	163.91	104%
	Ni^{2+} $2p_{3/2}$	852.97	
	Ni^{2+} $2p_{1/2}$	870.27	
	Ni^{2+} $2p_{3/2,\ sat.}$	854.27	
Ni	Ni^{2+} $2p_{1/2,\ sat.}$	874.26	100%



Supplementary Figure 14. XPS spectra of Co-MFU-4*l*-Cl: (a) survey, (b) C1s, (c)O1s, (d)N1s, (e) Ni2p and (f) Cl2p.

Supplementary Table 10. Summary of XPS spectra peaks of Co-MFU-4*l*-Cl

Element	Chemical State	Bonding Energy(eV)	Atomic % per 1/5 metal
	C=C	284.06	
C	C=N	285.43	
	H_2O (coordinates to Co)/O-C	531.39	
O	H_2O (in cavity)	533.15	
N	N=C/N=N	399.55	
	Cl^- $2p_{3/2}$	197.86	
Cl	Cl^- $2p_{1/2}$	199.44	488%
	Co^{2+} $2p_{3/2}$	781.18	
	Co^{2+} $2p_{1/2}$	796.87	
	Co^{2+} $2p_{3/2, sat.}$	786.21	
Co	Co^{2+} $2p_{1/2, sat.}$	802.70	384%
	Zn^{2+} $2p_{3/2}$	1022.25	
Zn	Zn^{2+} $2p_{1/2}$	1045.35	116%

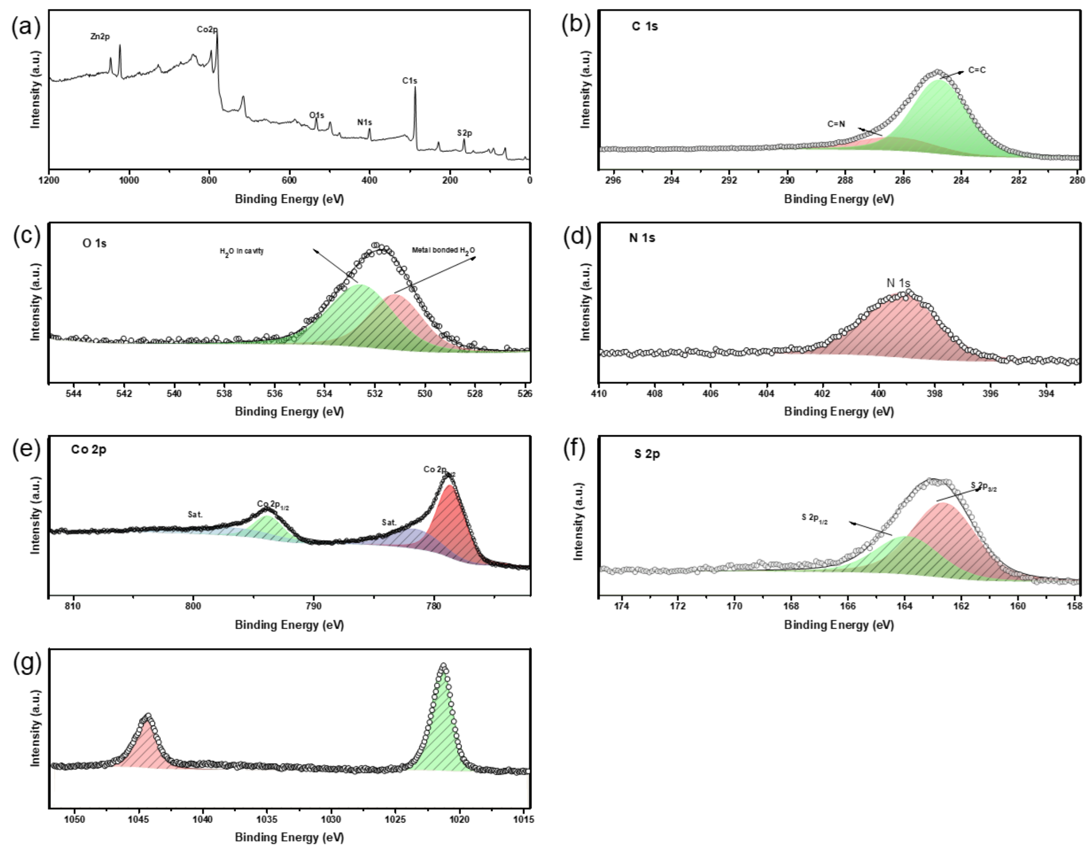


Supplementary Figure 15. XPS spectra of Co-MFU-4*l*-OH: (a) survey, (b) C1s, (c)O1s, (d)N1s, (e) Ni2p and (f) Cl2p.

Supplementary Table 11. Summary of XPS spectra peaks of Co-MFU-4*l*-OH

Element	Chemical State	Bonding Energy(eV)	Atomic % per 1/5 metal
	C=C	284.69	
	C=N	285.95	
C	C-O	288.66	
	-OH (bridging)	530.30	
	H_2O (coordinates to Co)/O-C	531.86	
O	H_2O (in cavity)	533.52	
N	N=C/N=N	399.55	
Cl	N/A	N/A	0%
	Co^{2+} $2p_{3/2}$	781.58	
	Co^{2+} $2p_{1/2}$	797.06	
	Co^{2+} $2p_{3/2, sat.}$	786.71	
Co	Co^{2+} $2p_{1/2, sat.}$	803.5	409%
Zn	Zn^{2+} $2p_{3/2}$	1022.29	91%

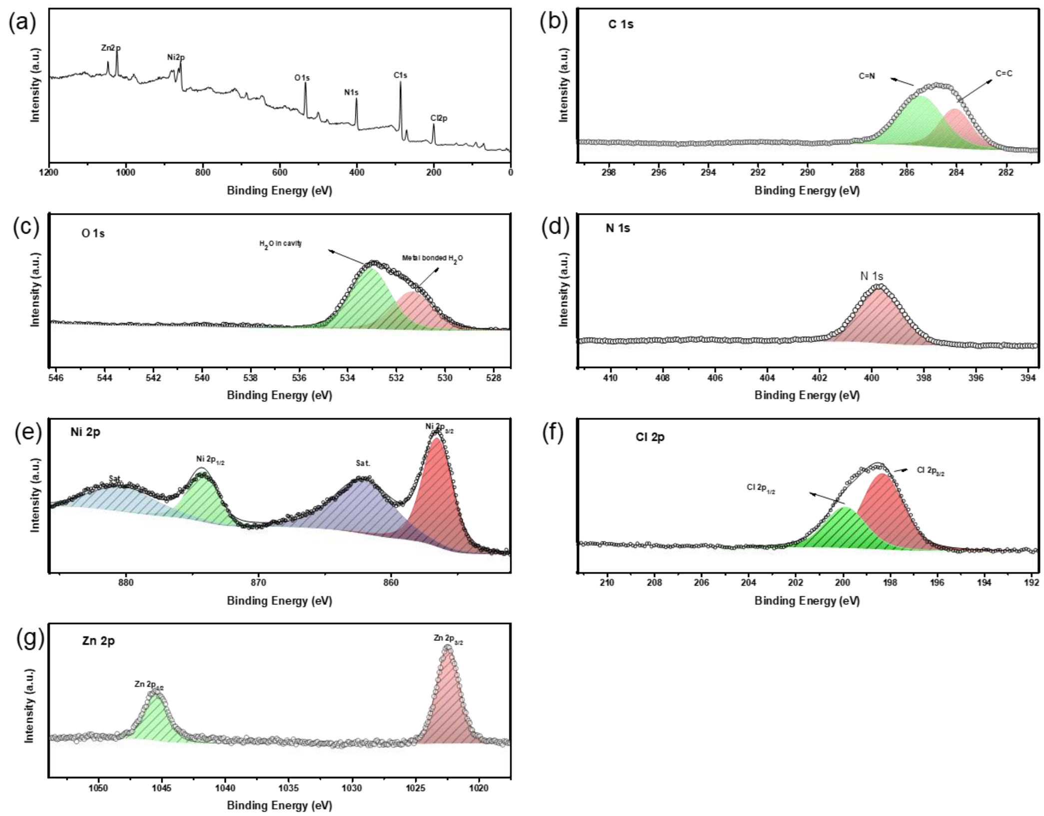
	Zn^{2+} $2p_{1/2}$	1045.35	



Supplementary Figure 16. XPS spectra of Co-MFU-4*l*-SH: (a) survey, (b) C1s, (c)O1s, (d)N1s, (e) Ni2p and (f) Cl2p.

Supplementary Table 12. Summary of XPS spectra peaks of Co-MFU-4*l*-SH

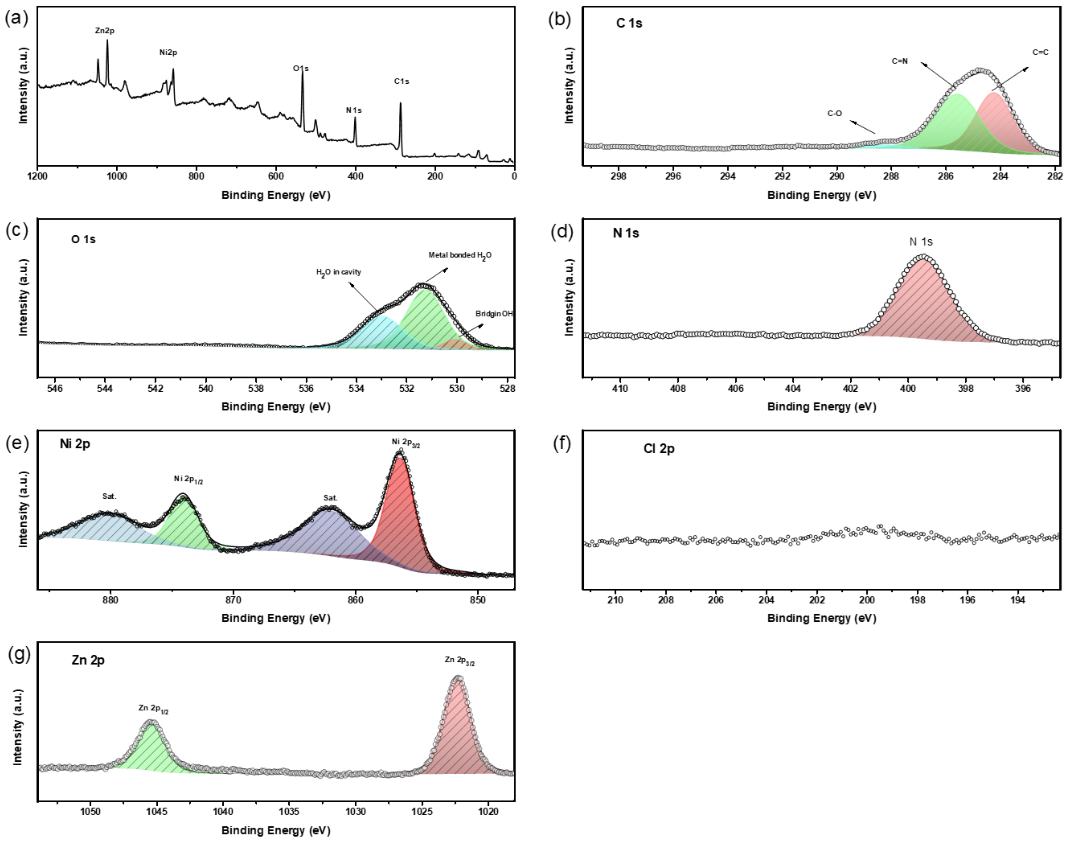
Element	Chemical State	Bonding Energy(eV)	Atomic % per 1/5 metal
	C=C	284.72	
C	C=N	286.25	
	H_2O (coordinates to Co)/O-C	531.14	
O	H_2O (in cavity)	532.52	
N	N=C/N=N	399.18	
	S^{2-} $2p_{3/2}$	162.57	
S	S^{2-} $2p_{1/2}$	163.87	367%
	Co^{2+} $2p_{3/2}$	778.55	
	Co^{2+} $2p_{1/2}$	793.71	
	Co^{2+} $2p_{3/2, sat.}$	781.5	
Co	Co^{2+} $2p_{1/2, sat.}$	796.66	399%
	Zn^{2+} $2p_{3/2}$	1022.25	
Zn	Zn^{2+} $2p_{1/2}$	1045.35	101%



Supplementary Figure 17. XPS spectra of Ni-MFU-4*l*-Cl: (a) survey, (b) C1s, (c)O1s, (d)N1s, (e) Ni2p and (f) Cl2p.

Supplementary Table 13. Summary of XPS spectra peaks of Ni-MFU-4*l*-Cl

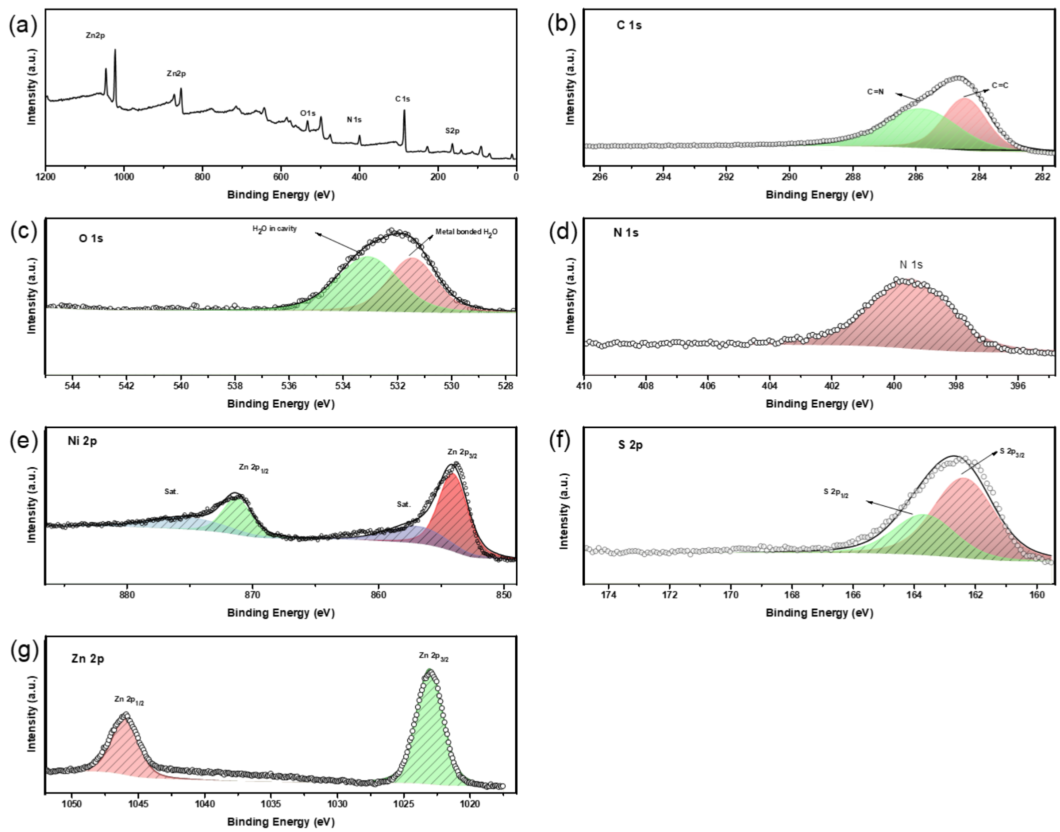
Element	Chemical State	Bonding Energy(eV)	Atomic % per 1/5 metal
	C=C	284.04	
C	C=N	285.41	
	H_2O (coordinates to Co)/O-C	531.33	
O	H_2O (in cavity)	533.05	
N	N=C/N=N	399.66	
	Cl^- $2p_{3/2}$	198.3	
Cl	Cl^- $2p_{1/2}$	199.88	377%
	Ni^{2+} $2p_{3/2}$	856.50	
	Ni^{2+} $2p_{1/2}$	874.17	
	Ni^{2+} $2p_{3/2,\ sat.}$	862.11	
Ni	Ni^{2+} $2p_{1/2,\ sat.}$	880.23	273%
	Zn^{2+} $2p_{3/2}$	1022.41	
Zn	Zn^{2+} $2p_{1/2}$	1045.41	227%



Supplementary Figure 18. XPS spectra of Ni-MFU-4*l*-OH: (a) survey, (b) C1s, (c)O1s, (d)N1s, (e) Ni2p and (f) Cl2p.

Supplementary Table 14. Summary of XPS spectra peaks of Ni-MFU-4*l*-OH

Element	Chemical State	Bonding Energy(eV)	Atomic % per 1/5 metal
	C=C	284.20	
	C=N	285.54	
C	C-O	288.27	
	-OH (bridging)	530.03	
	H_2O (coordinates to Co)/O-C	531.22	
O	H_2O (in cavity)	532.94	
N	N=C/N=N	399.41	
Cl	N/A	N/A	0%
	Ni^{2+} $2p_{3/2}$	856.29	
	Ni^{2+} $2p_{1/2}$	873.99	
	Ni^{2+} $2p_{3/2,\ sat.}$	861.99	
Ni	Ni^{2+} $2p_{1/2,\ sat.}$	880.05	305%
	Zn^{2+} $2p_{3/2}$	1022.29	
Zn	Zn^{2+} $2p_{1/2}$	1045.35	195%

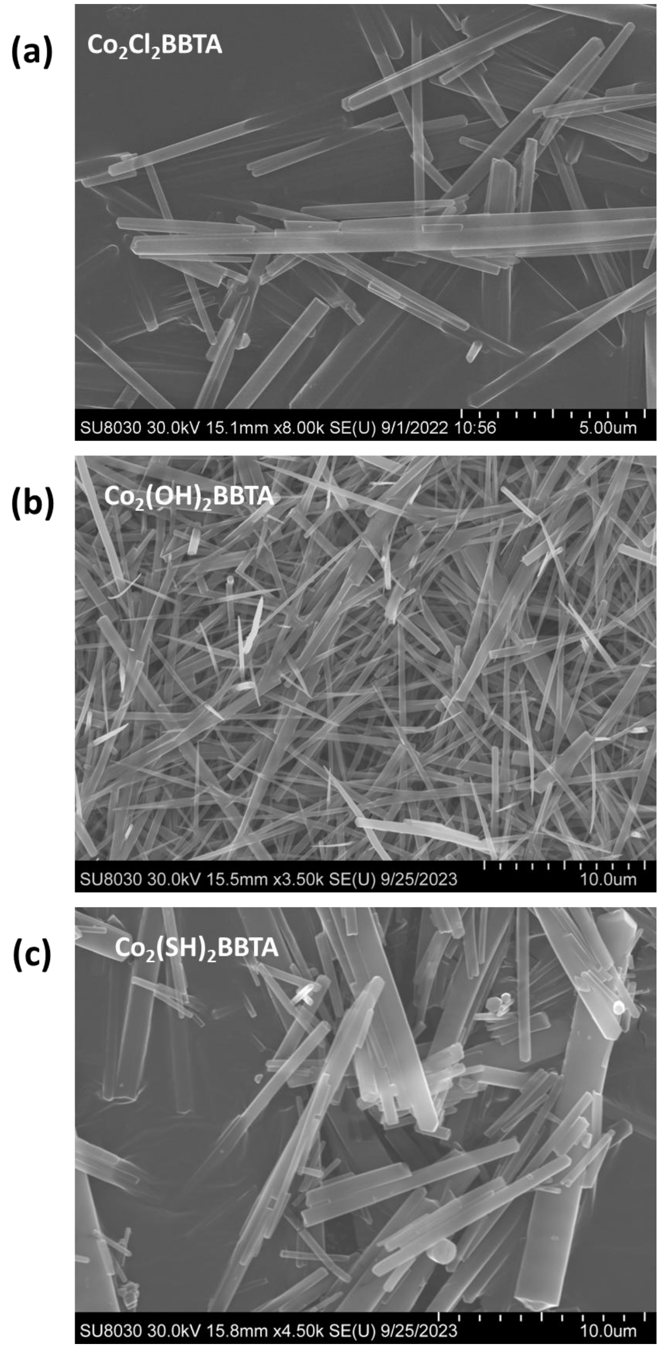


Supplementary Figure 19. XPS spectra of Ni-MFU-4*l*-SH: (a) survey, (b) C1s, (c)O1s, (d)N1s, (e) Ni2p and (f) Cl2p.

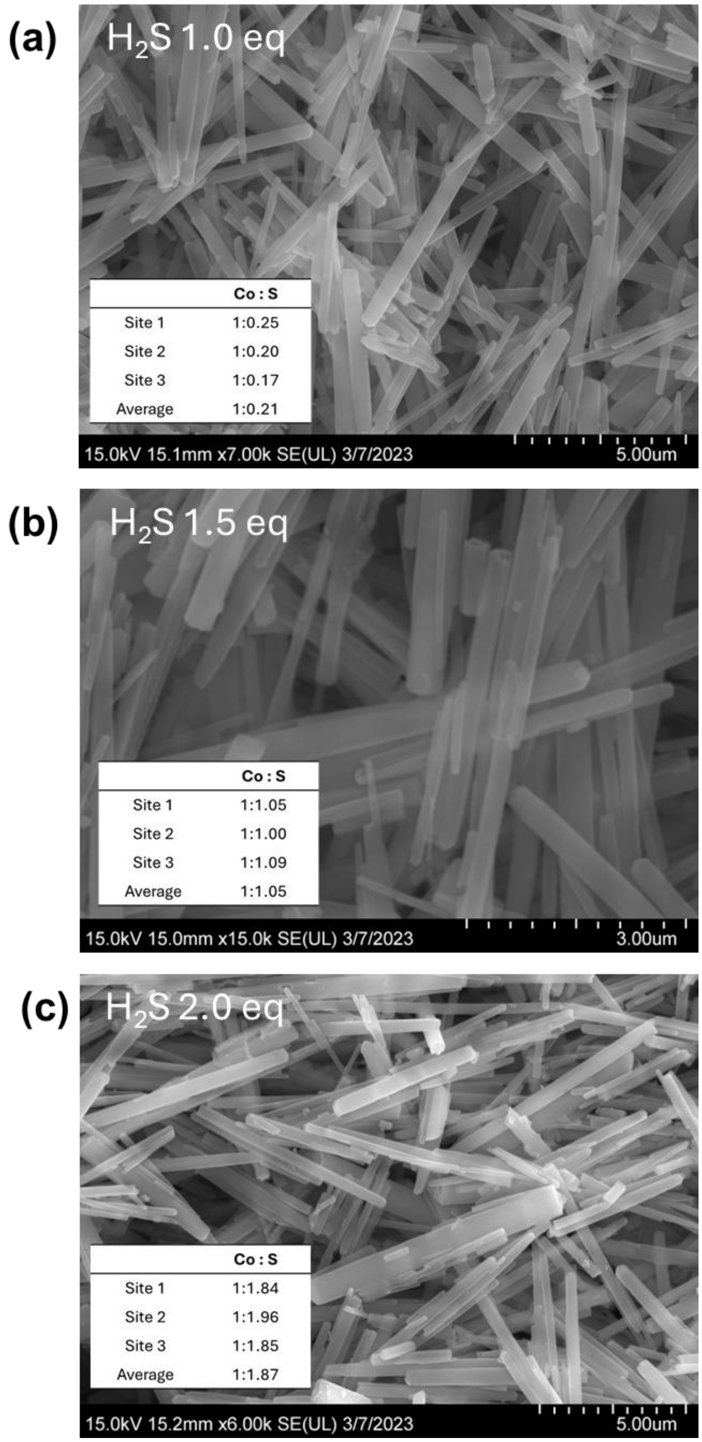
Supplementary Table 15. Summary of XPS spectra peaks of Ni-MFU-4*l*-SH

Element	Chemical State	Bonding Energy(eV)	Atomic % per 1/5 metal
	C=C	284.42	
C	C=N	285.82	
	H_2O (coordinates to Co)/O-C	531.43	
O	H_2O (in cavity)	533.07	
N	N=C/N=N	399.46	
	S^{2-} $2p_{3/2}$	162.35	
S	S^{2-} $2p_{1/2}$	163.65	294%
	Ni^{2+} $2p_{3/2}$	854.01	
	Ni^{2+} $2p_{1/2}$	871.24	
	Ni^{2+} $2p_{3/2,\ sat.}$	856.83	
Ni	Ni^{2+} $2p_{1/2,\ sat.}$	875.16	212%
	Zn^{2+} $2p_{3/2}$	1023.01	
Zn	Zn^{2+} $2p_{1/2}$	1045.96	288%

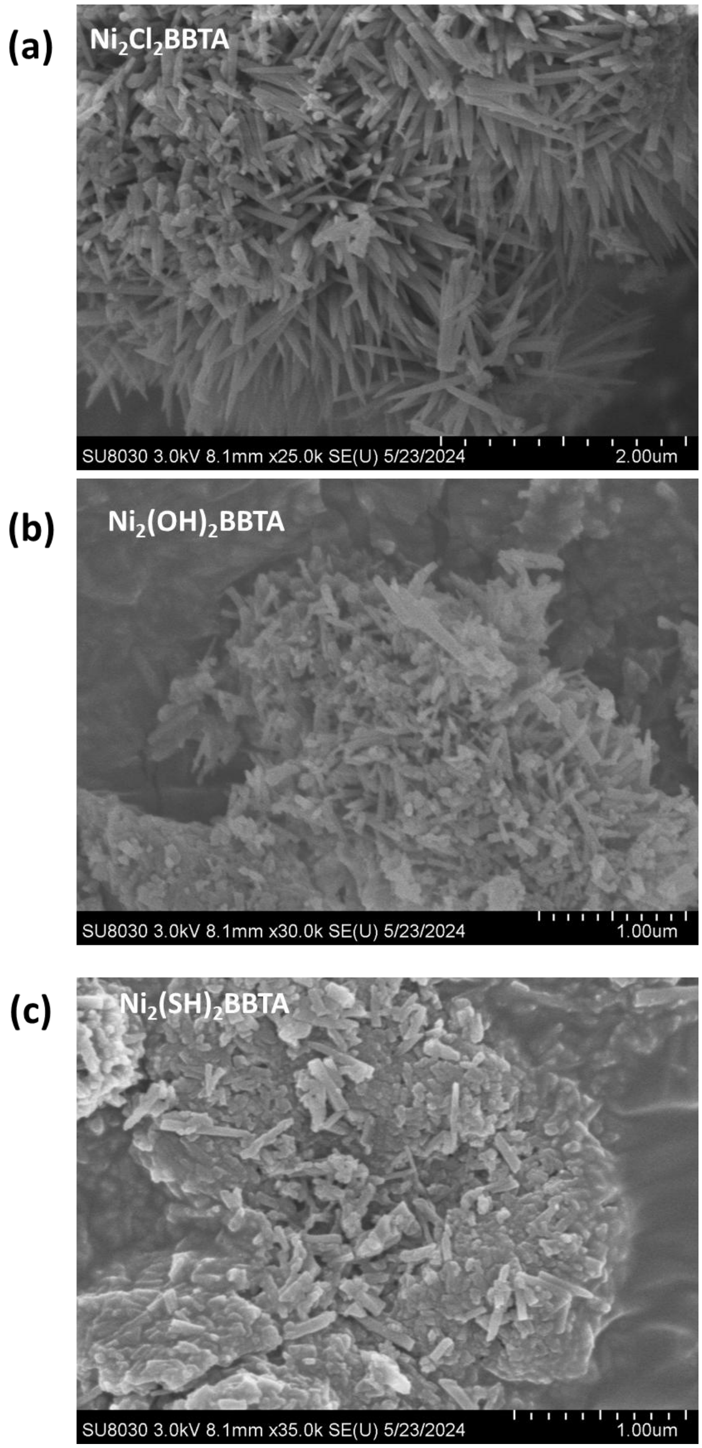
Section F. Scanning Electron Microscopy (SEM) and Energy Dispersive X-ray spectroscopy (EDS)



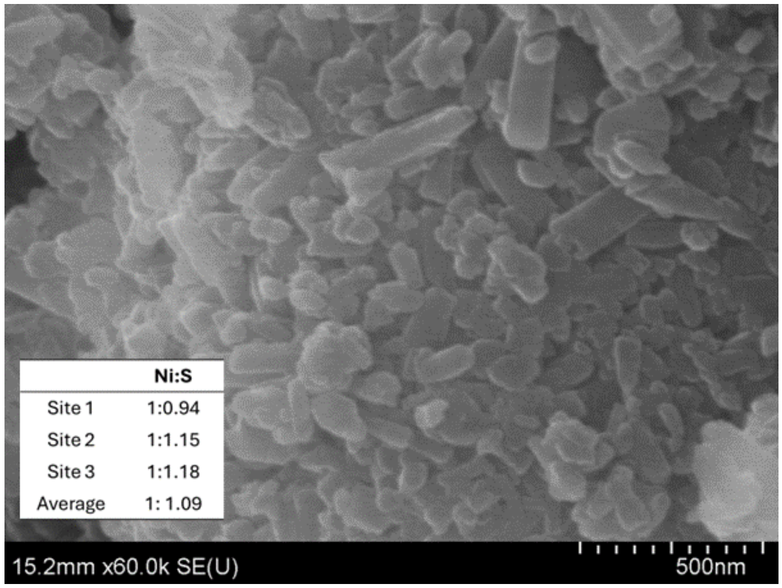
Supplementary Figure 20. SEM images of (a) pristine Co_2Cl_2BBTA and post-synthetically converted (b) $Co_2(OH)_2$BBTA, (c) $Co_2(SH)_2$BBTA.



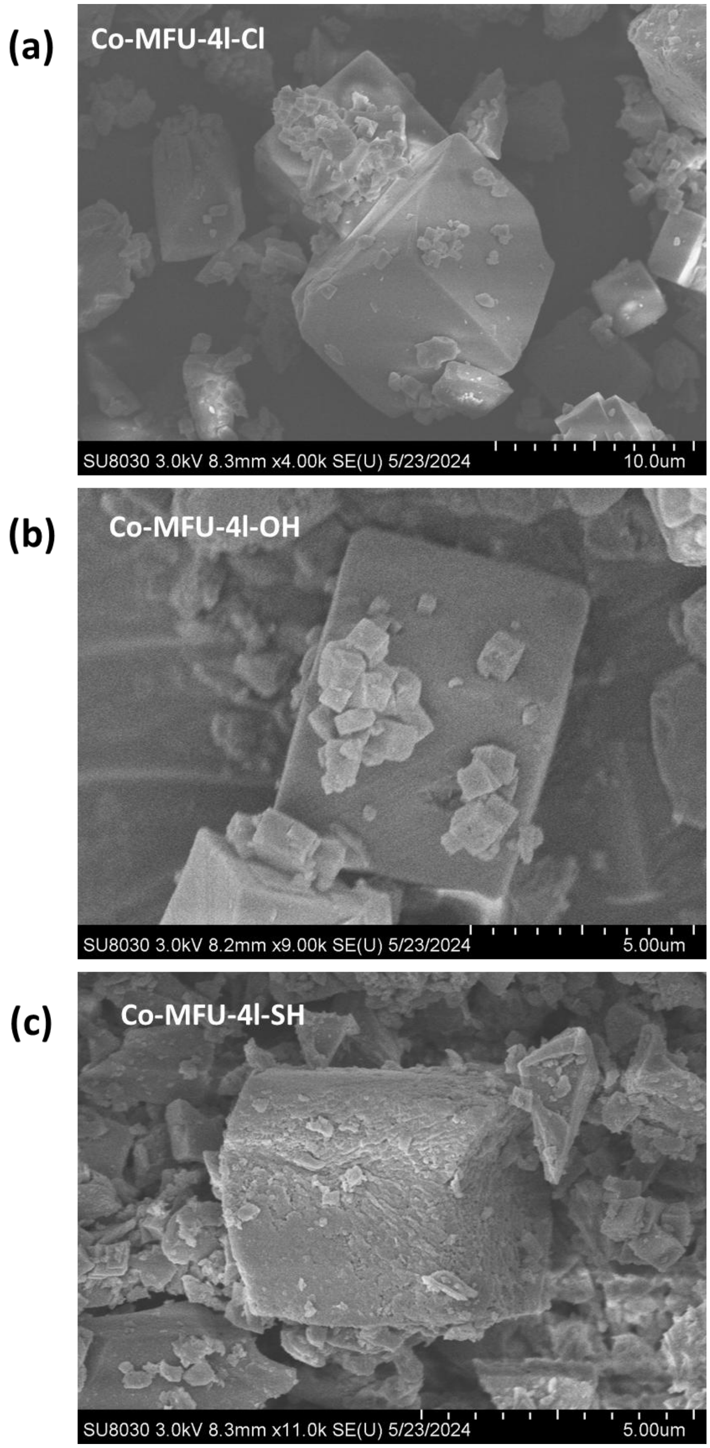
Supplementary Figure 21. SEM and EDS results of Co:S ratio of $Co_2(SH)_2BBTA$ with different equivalence (a) 1.0 eq. (b) 1.5 eq and (c) 2.0 eq of H_2S used.



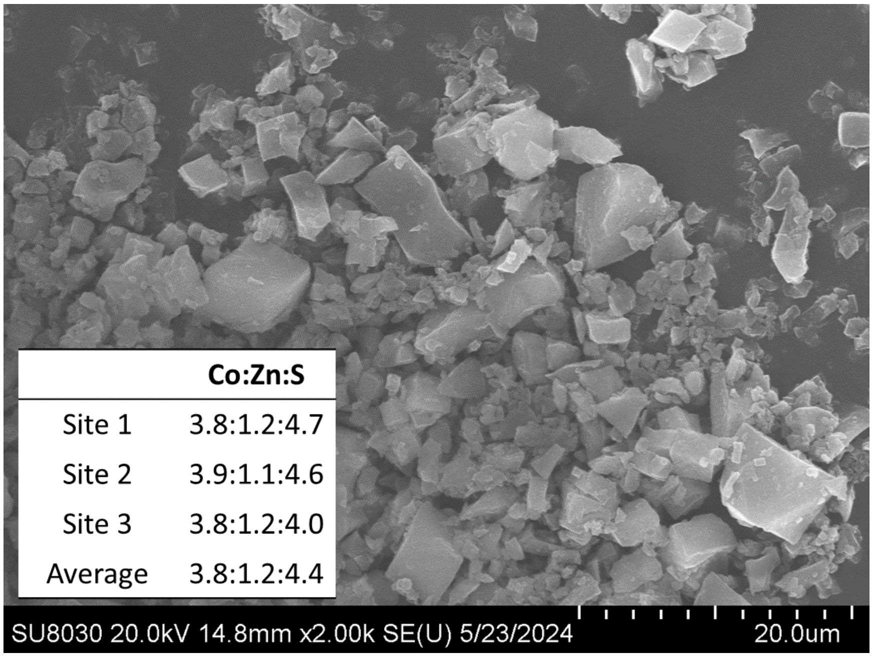
Supplementary Figure 22. SEM images of (a) pristine Ni_2Cl_2BBTA and post-synthetically converted (b) $Ni_2(OH)_2BBTA$, (c) $Ni_2(SH)_2BBTA$.



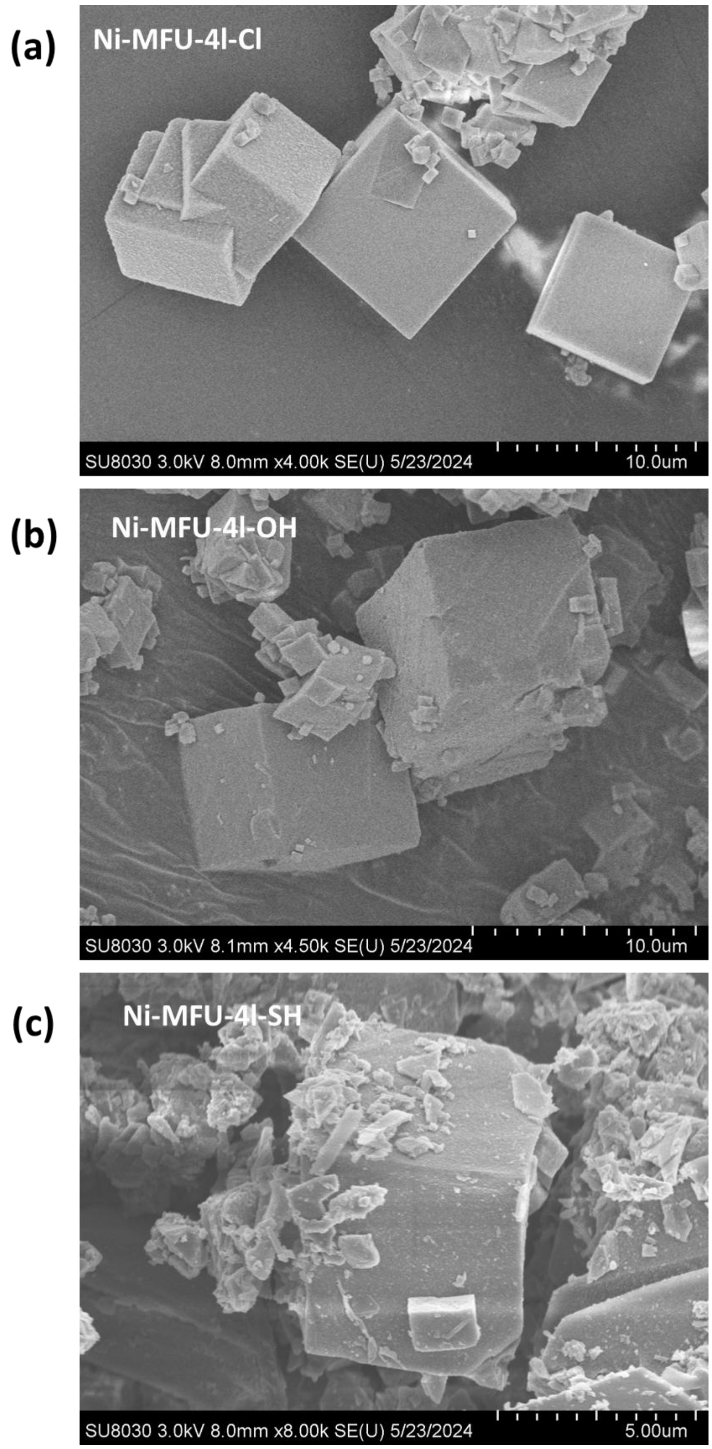
Supplementary Figure 23. SEM and EDS results of Ni:S ratio of $Ni_2(SH)_2$BBTA



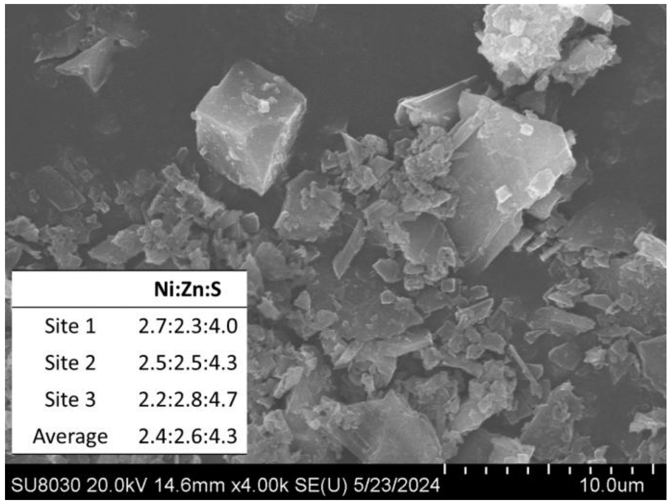
Supplementary Figure 24. SEM images of (a) pristine Co-MFU-4*l*-Cl and post-synthetically converted (b) Co-MFU-4*l*-OH, (c) Co-MFU-4*l*-SH.



Supplementary Figure 25. SEM and EDS results of Co:Zn:S ratio of Co-MFU-4*l*-SH



Supplementary Figure 26. SEM images of (a) pristine Ni-MFU-4*l*-Cl and post-synthetically converted (b) Ni-MFU-4*l*-OH, (c) Ni-MFU-4*l*-SH.

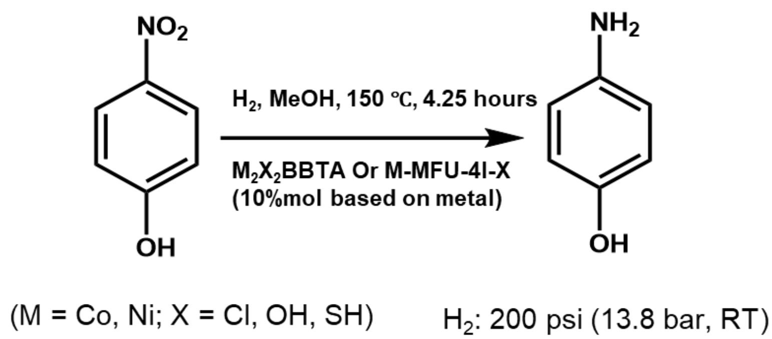


	Ni:Zn:S
Site 1	2.7:2.3:4.0
Site 2	2.5:2.5:4.3
Site 3	2.2:2.8:4.7
Average	2.4:2.6:4.3

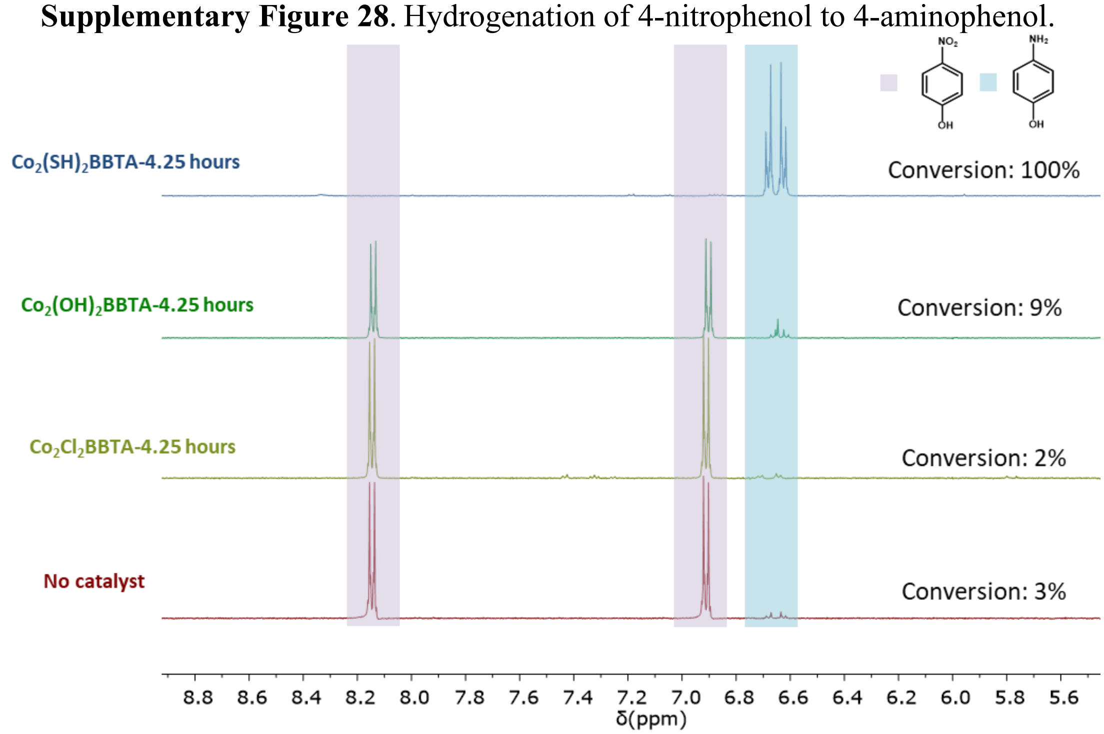
Supplementary Figure 27. SEM and EDS results of Ni:Zn:S ratio of Ni-MFU-4*l*-SH

Section G. Hydrogenation Catalysis of 4-Nitrophenol

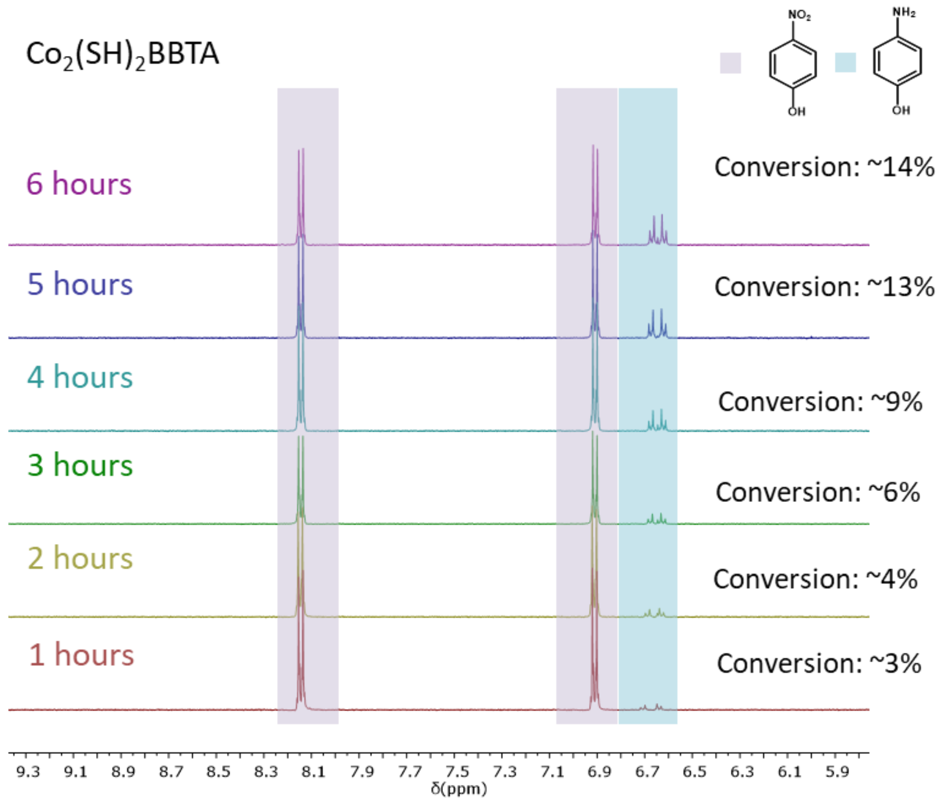
The experiment was conducted using a Parr reactor. Initially, 20 mg of 4-nitrophenol was dissolved in 6 mL of methanol (MeOH) within a 25 mL Teflon liner. The MOFs were added in an amount corresponding to 10% based on the metal sites. The reactor system was purged with argon at 10 bar pressure three times, followed by purging with hydrogen gas at 13.8 bar pressure three times. The final pressure of hydrogen gas was kept at 13.8 bar at room temperature. Subsequently, the mixture was heated to 150°C (No significant changes in the mass of the MOFs were observed at the reaction temperature, as confirmed by TGA analysis (Supplementary Figure 55).) for 4.25 hours and then allowed to cool to room temperature. The product was separated using a centrifuge, and the conversion rate was monitored via NMR spectroscopy. The turnover frequency measurements were formed with 1.5 mol% $Co_2(SH)_2$BBTA and 5 mol% Ni-MFU-4*l*-SH.



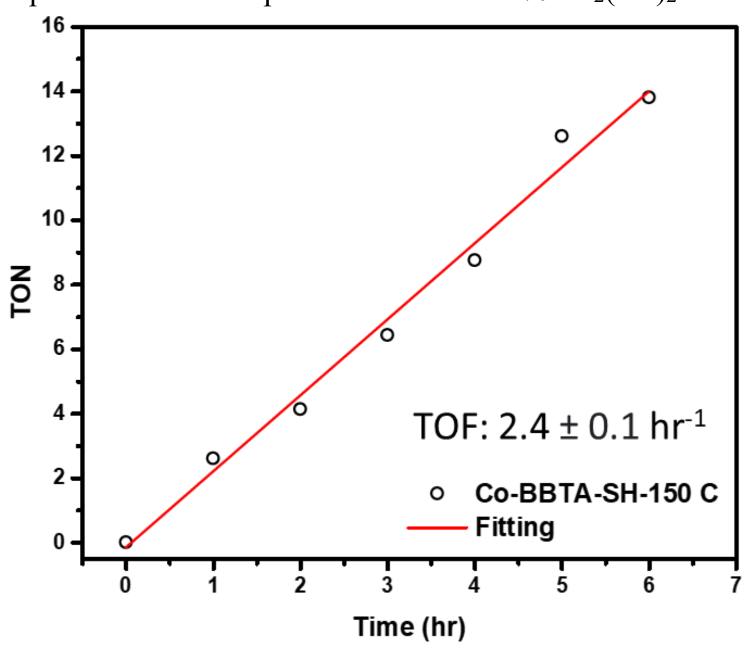
Supplementary Figure 28. Hydrogenation of 4-nitrophenol to 4-aminophenol.



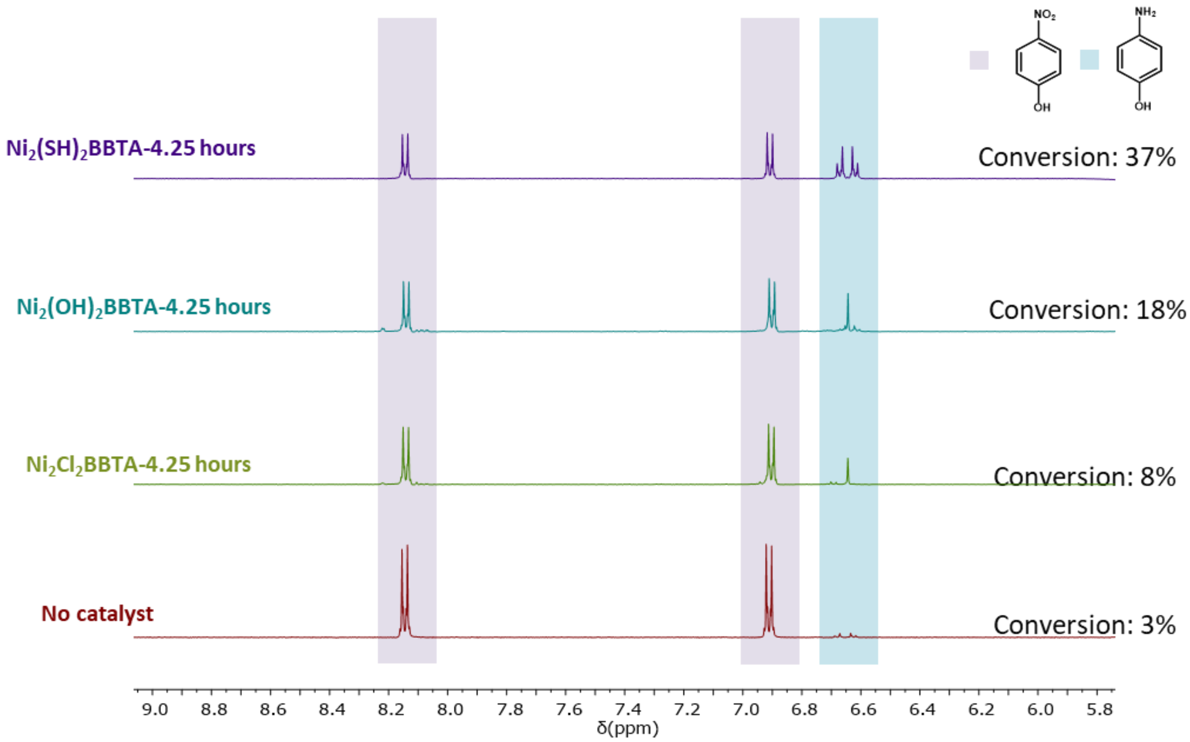
Supplementary Figure 29. ^{1}H NMR (500 MHz in methanol-d4, 298 K) of hydrogenation of 4-nitrophenol to 4-aminophenol with Co_2X_2BBTA as catalysts, where X = Cl, OH and SH.



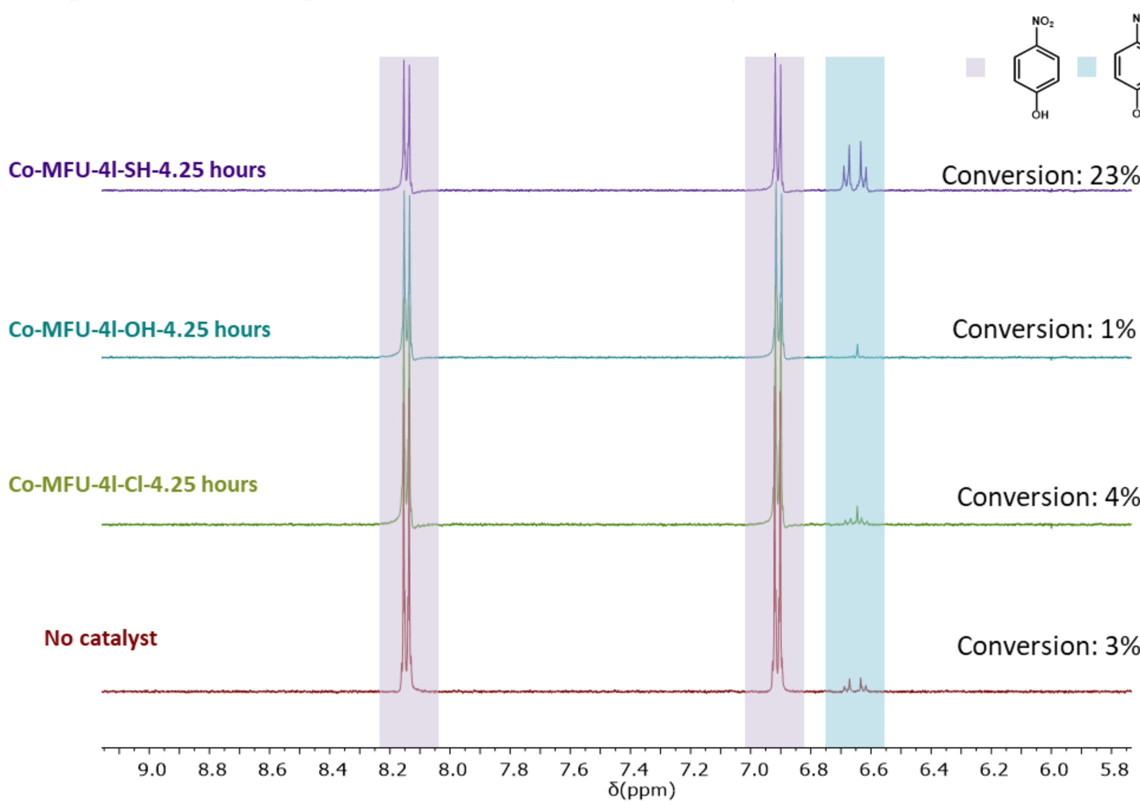
Supplementary Figure 30. ^1H NMR (500 MHz in methanol-d4, 298 K) of hydrogenation of 4-nitrophenol to 4-aminophenol with 1.5 mol% $Co_2(SH)_2BBTA$



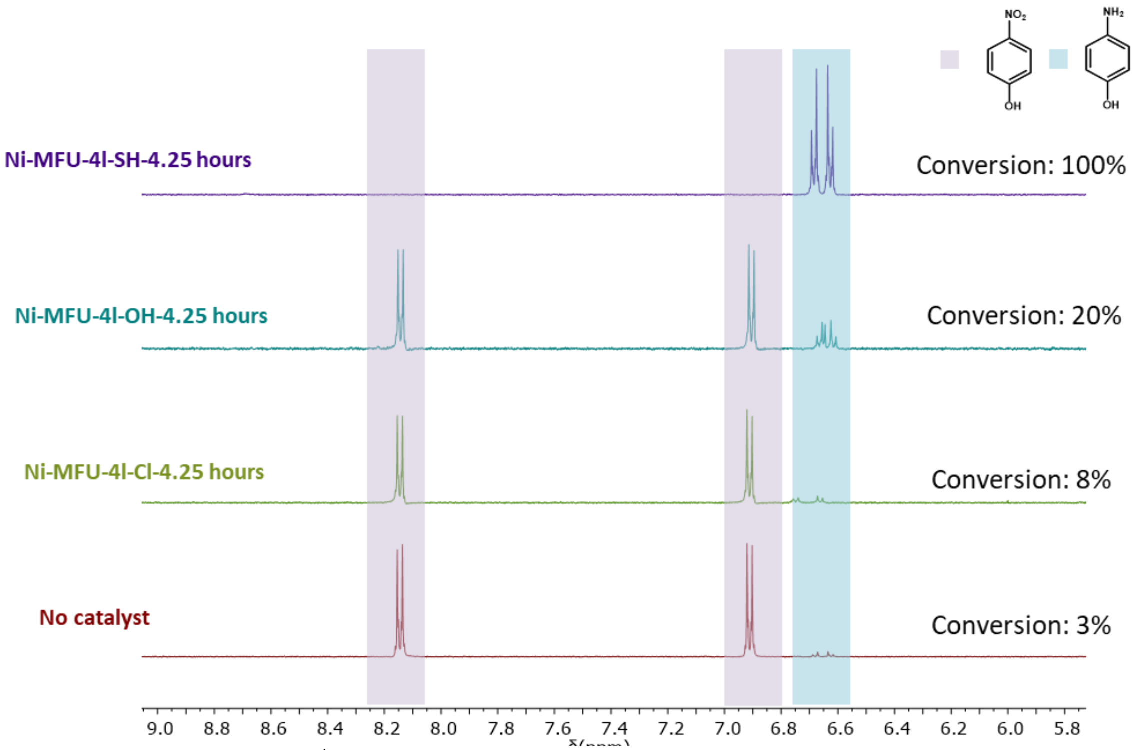
Supplementary Figure 31. Turnover frequency of hydrogenation of 4-nitrophenol to 4-aminophenol with 1.5 mol% $Co_2(SH)_2BBTA$



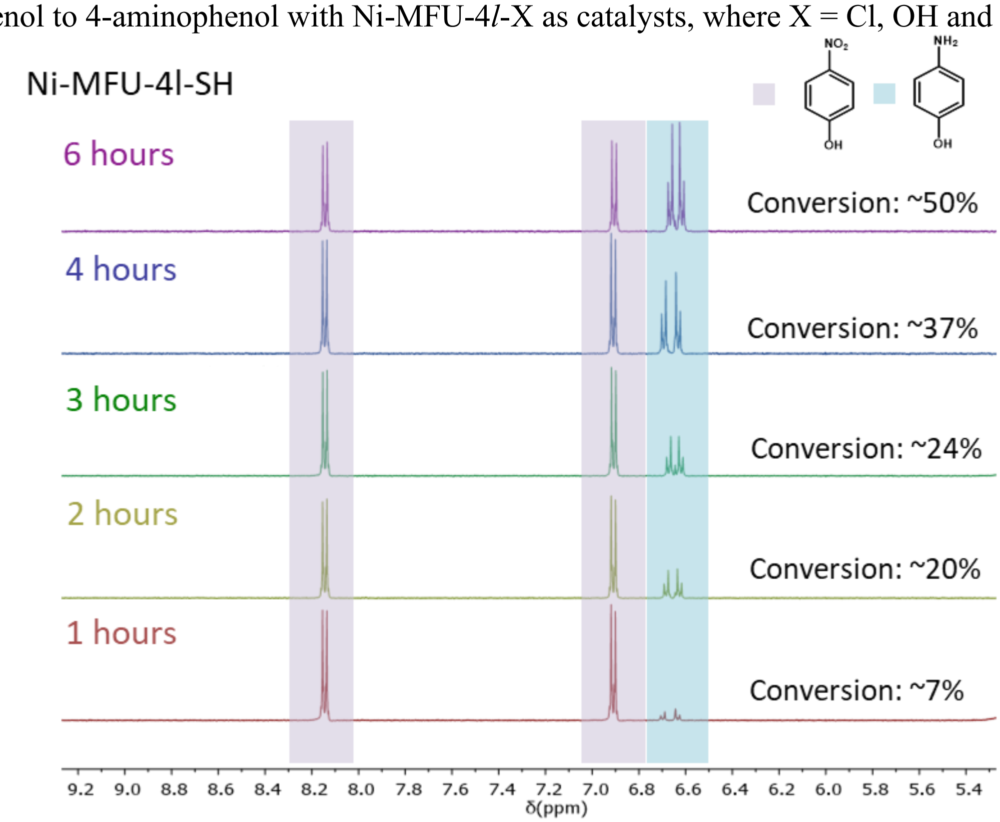
Supplementary Figure 32. [1]H NMR (500 MHz in methanol-4d, 298 K) of hydrogenation of 4-nitrophenol to 4-aminophenol with Ni_2X_2BBTA as catalysts, where X = Cl, OH and SH.



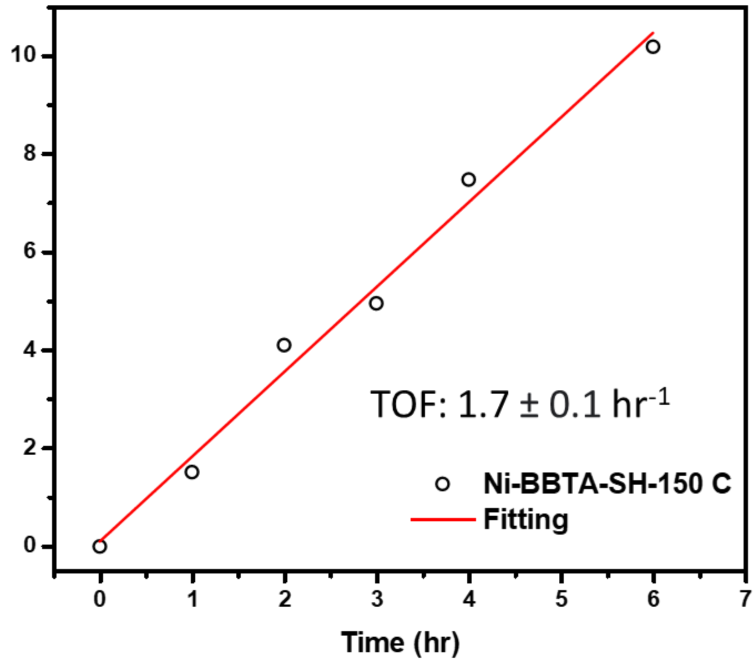
Supplementary Figure 33. [1]H NMR (500 MHz in methanol-d4, 298 K) of hydrogenation of 4-nitrophenol to 4-aminophenol with Co-MFU-4*l*-X as catalysts, where X = Cl, OH and SH.



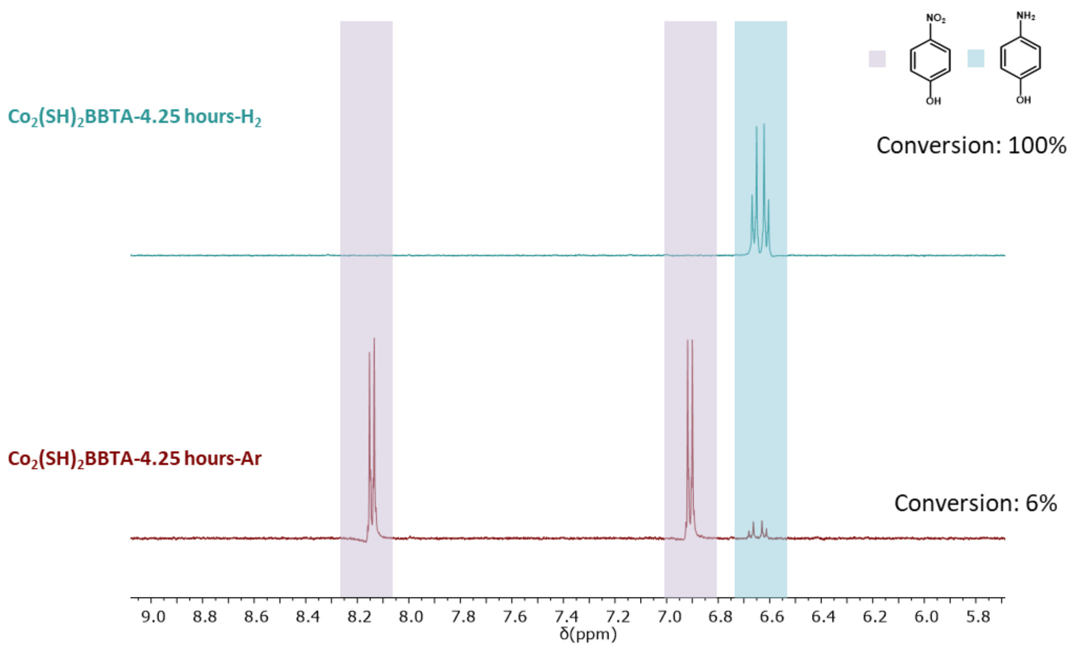
Supplementary Figure 34. [1]H NMR (500 MHz in methanol-d4, 298 K) of hydrogenation of 4-nitrophenol to 4-aminophenol with Ni-MFU-4*l*-X as catalysts, where X = Cl, OH and SH.



Supplementary Figure 35. [1]H NMR (500 MHz in methanol-d4, 298 K) of hydrogenation of 4-nitrophenol to 4-aminophenol with 5 mol% Ni-MFU-4*l*-SH



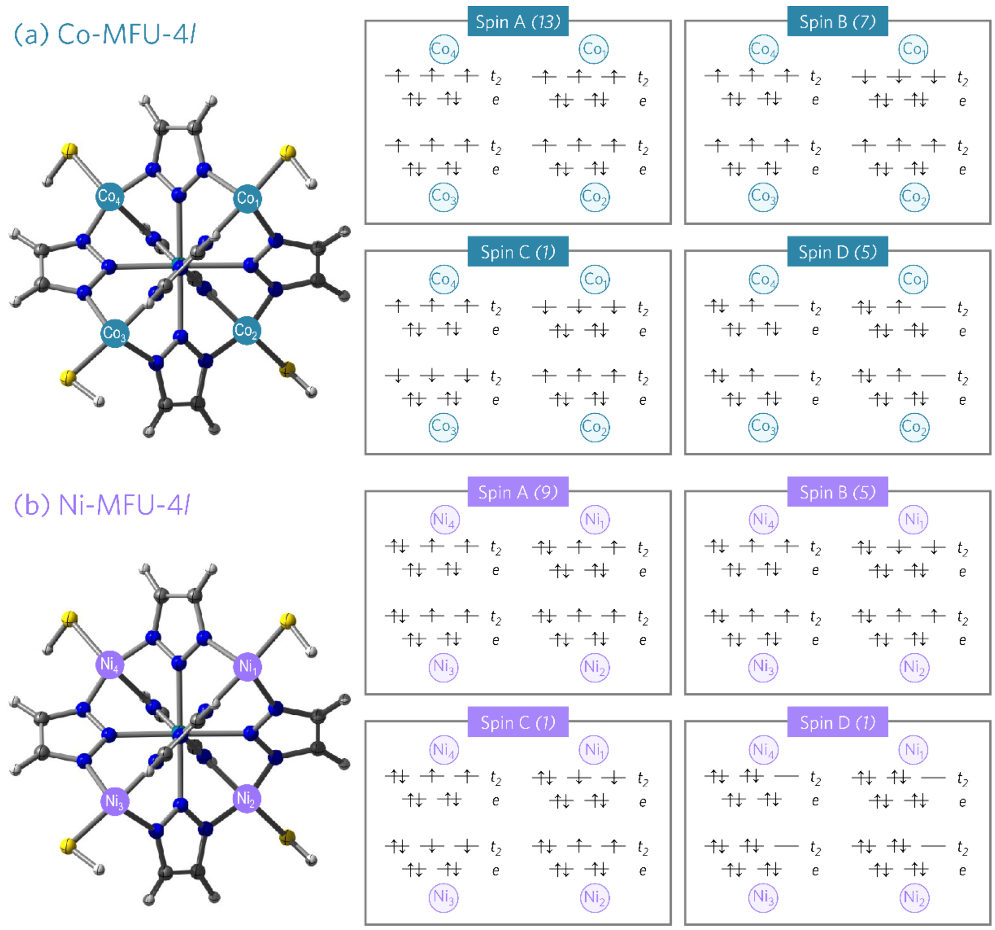
Supplementary Figure 36. Turnover frequency of hydrogenation of 4-nitrophenol to 4-aminophenol with 5 mol% Ni-MFU-4*l*-SH



Supplementary Figure 37. [1]H NMR (500 MHz in methanol-d4, 298 K) of hydrogenation of 4-nitrophenol to 4-aminophenol with 10 mol% $Co_2(SH)_2BBTA$ under Ar and H_2

Section H. Computational Analysis

H.1. 3D MFU-4*l* System: Spin Ladder

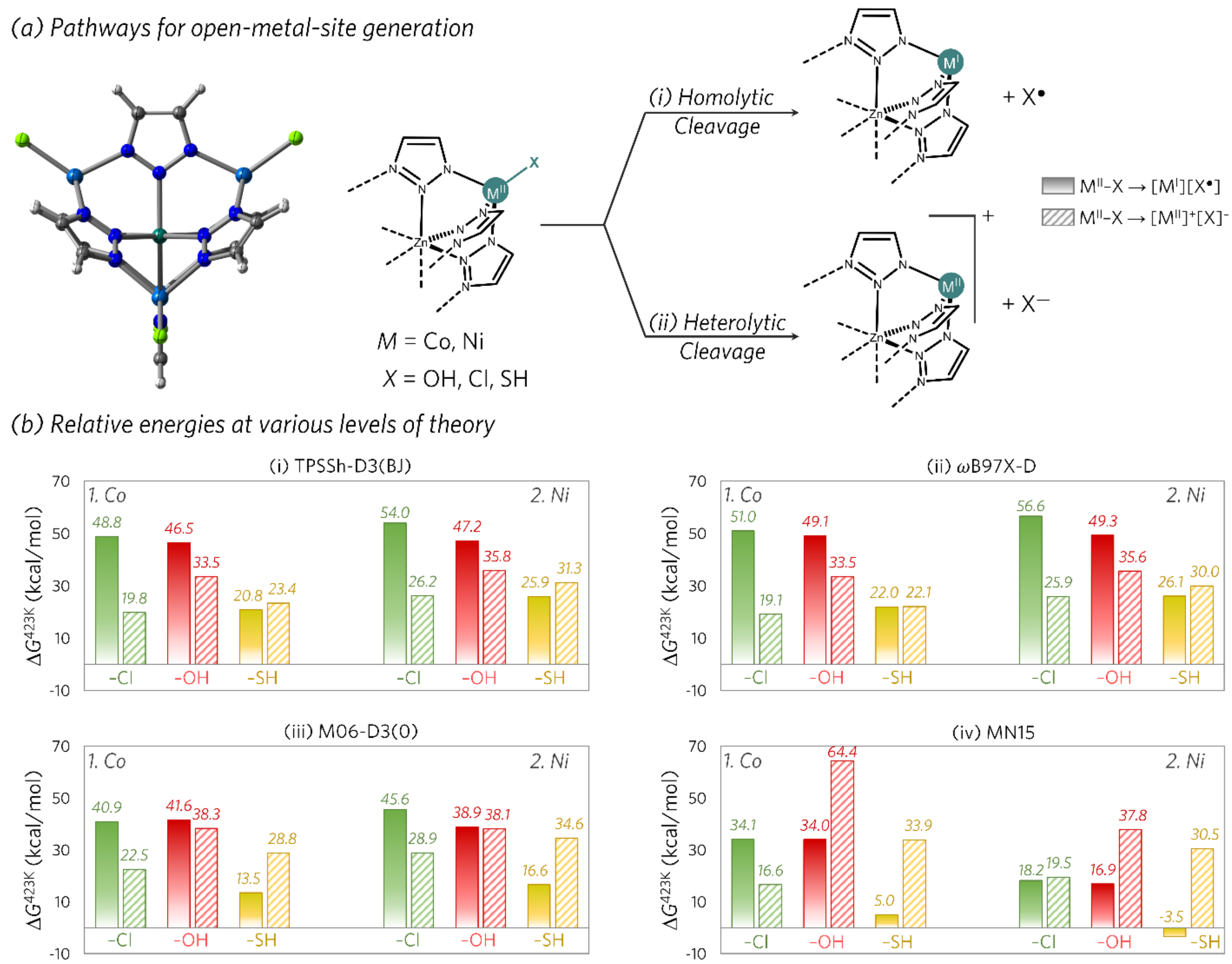


Supplementary Figure 38. Possible spin configurations for each metal center *M* (= Co^{II}/Ni^{II}) and their ferromagnetic and antiferromagnetic coupling combinations were explored thoroughly to understand the resting state electronic configuration in these systems. For each spin state, the total spin multiplicity of the node is shown in parenthesis.

Supplementary Table 16. Relative energies of the spin isomers presented in Supplementary Figure 38. Electronic energies (ΔE) are in kcal/mol calculated at the TPSSh-D3(BJ)/basis-II/SMD(MeOH)//TPSSh-D3(BJ)/basis-I level of theory.

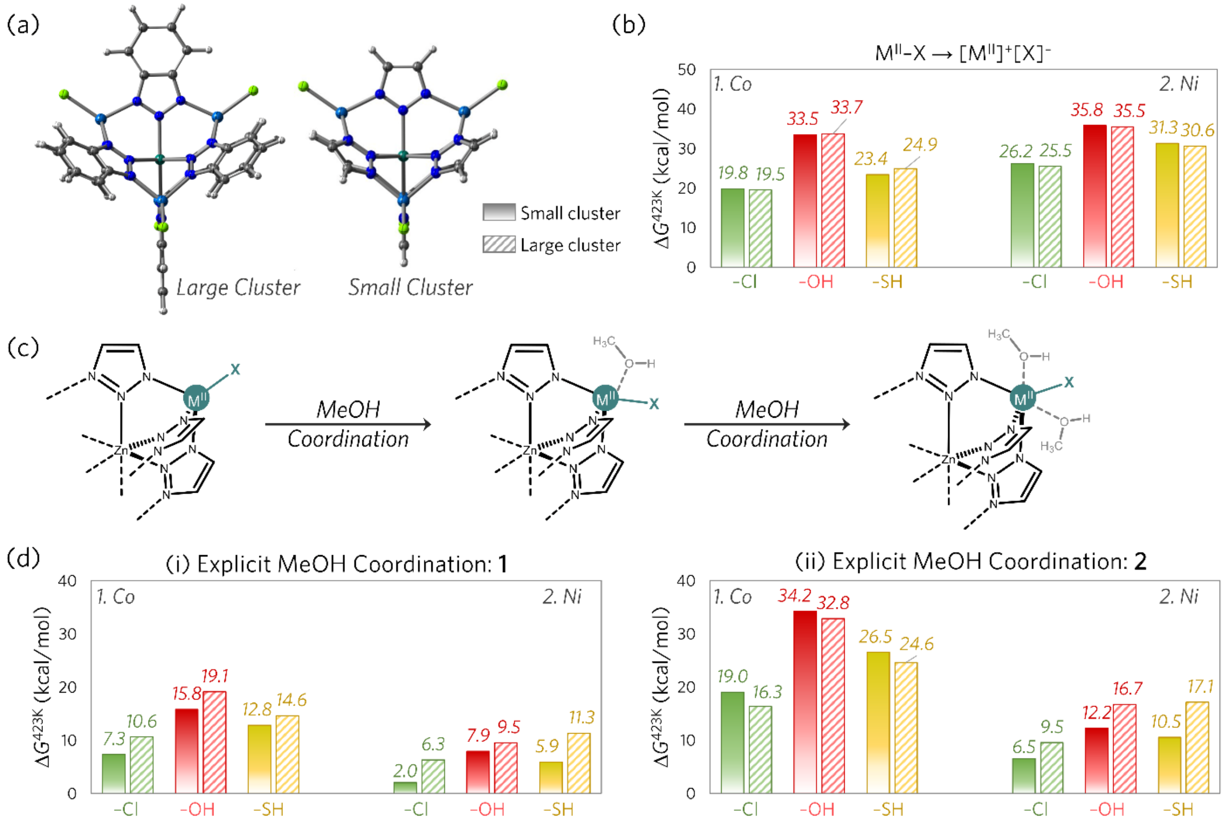
System	–X	ΔE (kcal/mol)			
		Spin-A	*Spin-B*	*Spin-C*	*Spin-D*
Co	–Cl	0	–0.2	–0.3	99.3
	–OH	0	–0.2	–0.3	70.2
	–SH	0	–0.3	–0.4	43.8
Ni	–Cl	0	–0.1	–0.1	64.3
	–OH	0	–0.2	–0.3	58.6
	–SH	0	–0.2	–0.1	58.2

H.2. 3D MFU-4*l* System: Choice of Density Functional



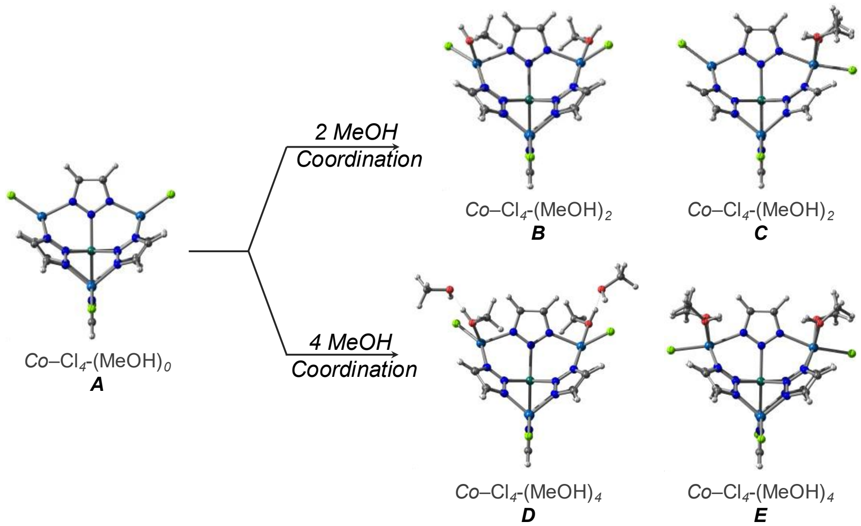
Supplementary Figure 39. (a) Schematic representation of the two possible pathways for open metal site generation: (i) homolytic cleavage and (ii) heterolytic cleavage of the M–X bond. (b) Influence of the density functional on the computed results. Four density functionals were employed: (i) TPSSh-D3(BJ), (ii) *ω*B97X-D,[2] (iii) M06-D3(0), and (iv) MN-15.[3] All methods showed a consistent trend: heterolytic M–X bond cleavage is preferred for *X* = Cl, OH (depicted by stripe filled bars), while for X = SH homolytic cleavage is preferred (depicted by solid gradient filled bar). For the rest of the study, results obtained using the M06-D3(0) functional will be shown. Gibbs free energies (at 150 °C) are given in kcal/mol.

H.3. Size of 3D Cluster Model: Small *vs.* Large

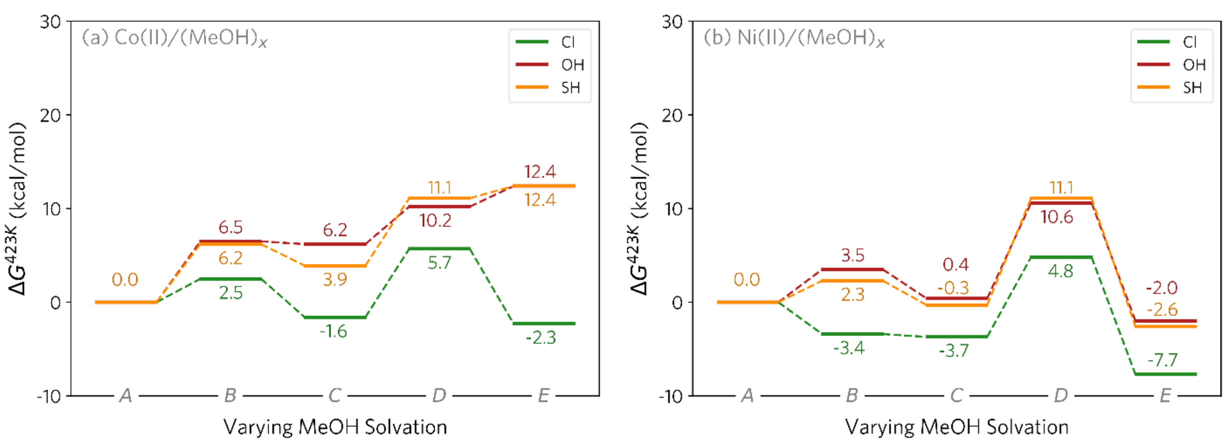


Supplementary Figure 40. Influence of the size the of the cluster model chosen for the computational study is analyzed. (a) "Large Cluster" (stripe filled bars) includes a benzotriazole ligand, while the "Small Cluster" (solid gradient filled bar) uses a triazole ligand. (b) Energetics of heterolytic M–X bond cleavage (schematically shown in Supplementary Figure 39(a)) for both cluster models. No significant difference is observed between the models. (c) and (d): Solvent effects were investigated using explicit solvation of the cluster models with methanol: (i) 1 MeOH per metal center and (ii) 2 MeOH per metal center. Panel (d) shows some variation in computed energetics for the larger cluster when compared against the smaller cluster, potentially due to steric hindrance. However, the relative trends for methanol solvation energies (M = Co, Ni; X = Cl, OH, SH) were consistent across both models. Based on these findings, subsequent results will be presented using the smaller cluster model only. Gibbs free energies (at 150 °C) are given in kcal/mol, calculated at the TPSSh-D3(BJ)/ def2-TZVP/SMD(MeOH)//TPSSh-D3(BJ)/def2-SVP level of theory.

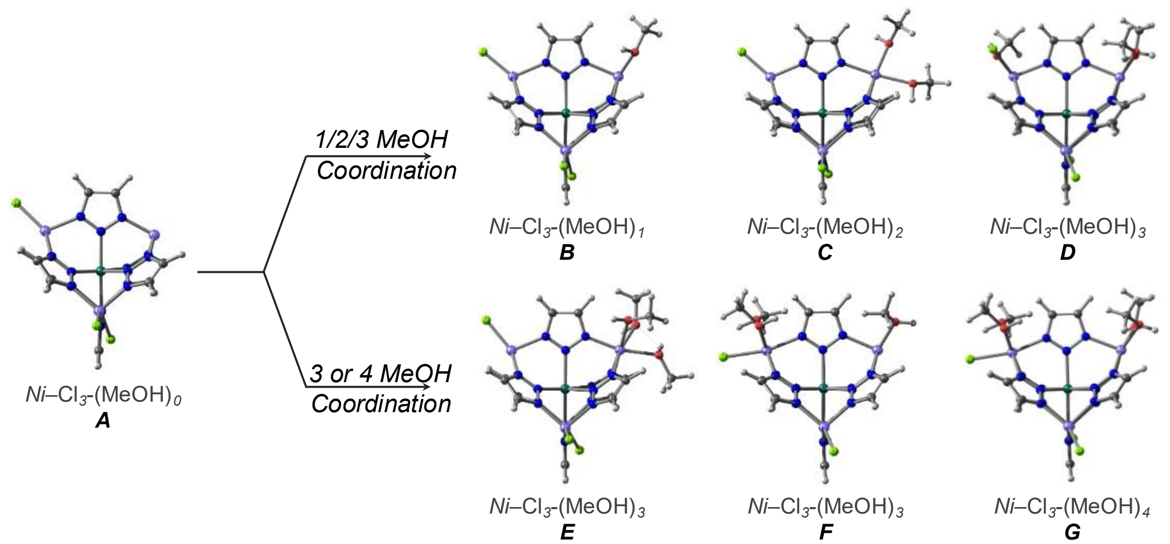
H.4. Effect of Systematic Methanol Coordination: 3D MOF Neutral M^{II}–X Species



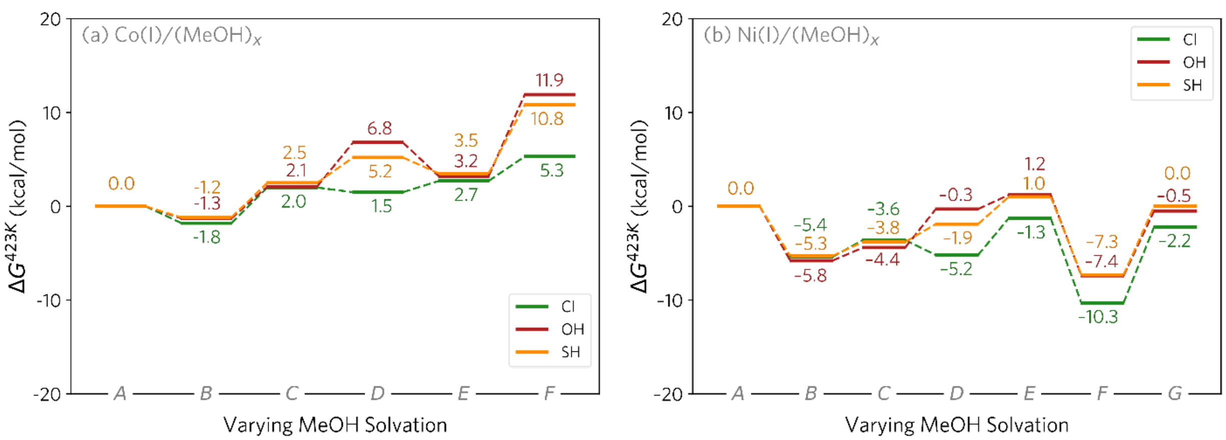
Supplementary Figure 41. Schematic representation of various coordination modes explored for MeOH binding to the neutral M(II)–X species: (A) Two tetrahedral M(II)–X sites (starting structure), (B) Two square pyramidal M(II) sites, (C) Combination of tetrahedral and octahedral M(II) sites, (D) Two square pyramidal M(II) sites with additional hydrogen bonded MeOH units (E) Two octahedral M(II) sites.



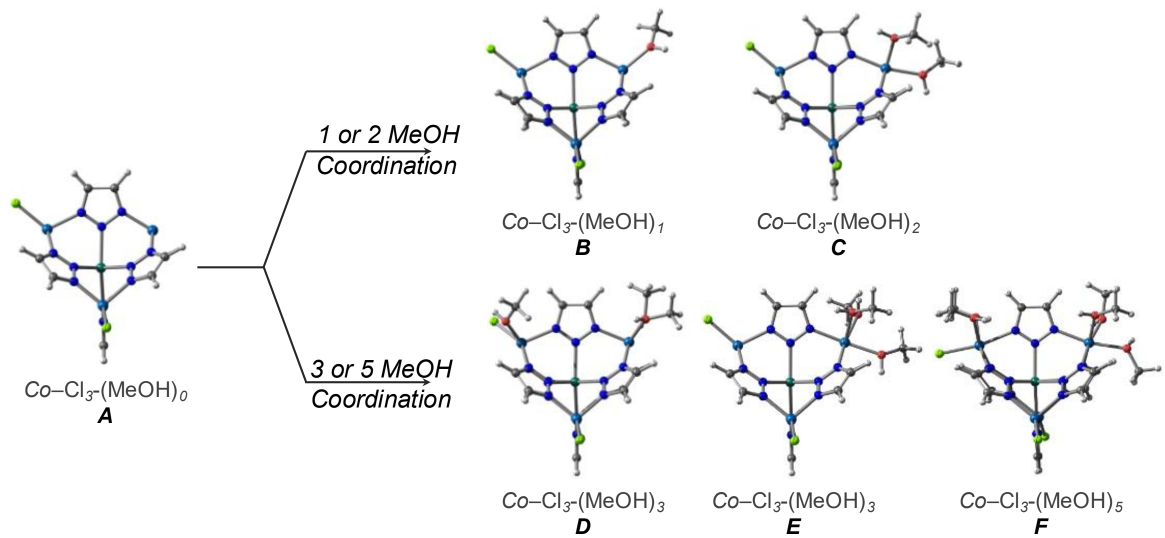
Supplementary Figure 42. MeOH coordination energetics for Co^{II} (a) and Ni^{II} (b) complexes. Ni consistently shows lower binding energies across all coordination modes. For Co at 423 K, most MeOH coordination modes are endergonic, with only two exceptions involving the Cl ligand (C and E). Thus, the unsolvated system is primarily relevant for neutral Co(II)–X species. In contrast, for Ni(II)–X (X = Cl, OH, SH), the octahedrally coordinated species E is considered the resting state. Gibbs free energies (at 150 °C) are given in kcal/mol, calculated at the M06-D3(0)/def2-TZVP/SMD(MeOH) //TPSSh-D3(BJ)/def2-SVP level of theory.



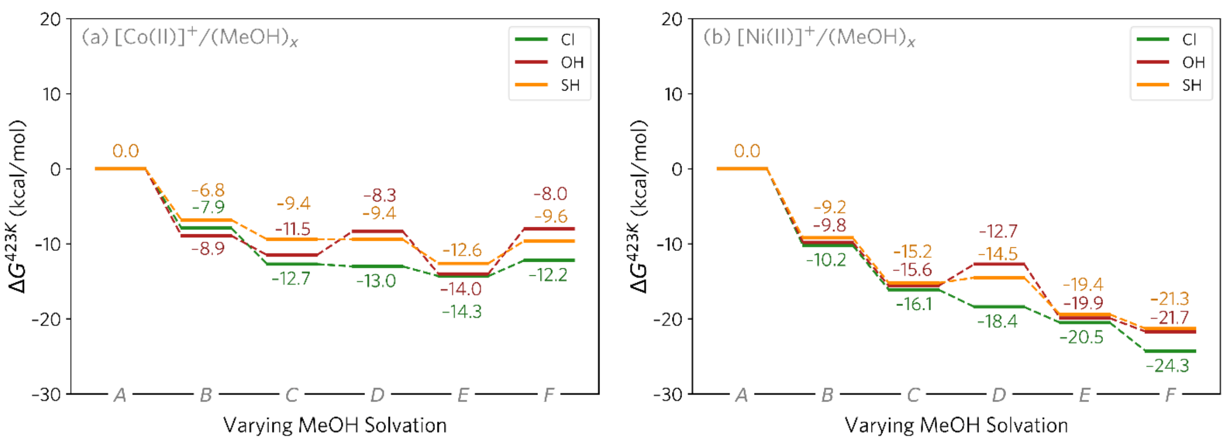
Supplementary Figure 43. Schematic representation of various coordination modes explored for MeOH binding to [M^I] sites: tetrahedral M^{II}–X and trigonal [M^I] sites (starting structure; **A**), and various other MeOH coordinated sites (**B** – **G**).



Supplementary Figure 44. Energetics of MeOH coordination at a [M(I)] species is shown for both Co (a) and Ni (b). Species **A** – **G** are demonstrated in Supplementary Figure 43 Overall, MeOH solvation at Ni is more favored compared to at Co for all coordination modes. At 423 K, the lowest energy structure for Co(I) is the solvated structure **B**, having one MeOH coordination at the open-metal site. For Ni(I), it is the solvated structure **F** that is analogous to **B**, having one MeOH coordination at the open-metal site, but also the other Ni center octahedrally coordinated by two more MeOH molecules. Gibbs free energies (at 150 °C) are given in kcal/mol, calculated at the M06-D3(0)/def2-TZVP/SMD(MeOH)//TPSSh-D3(BJ)/def2-SVP level of theory.

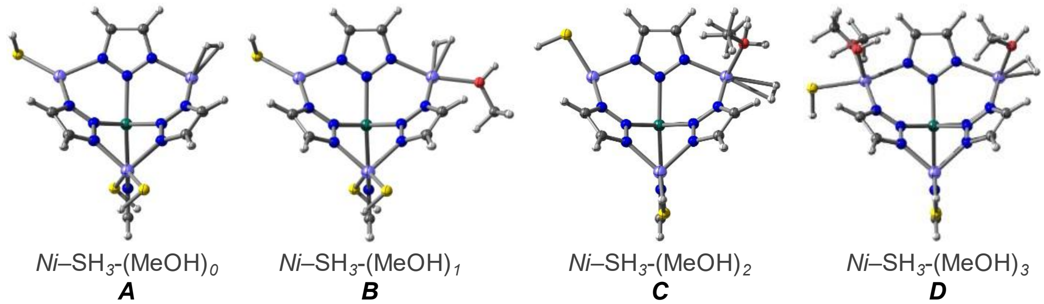


***Supplementary Figure 45**. Schematic representation of various coordination modes explored for MeOH binding to $[M^{II}]^+$ sites: tetrahedral M^{II}–X and trigonal $[M^{II}]^+$ sites (starting structure; A), and various other MeOH coordinated sites (B – F).*

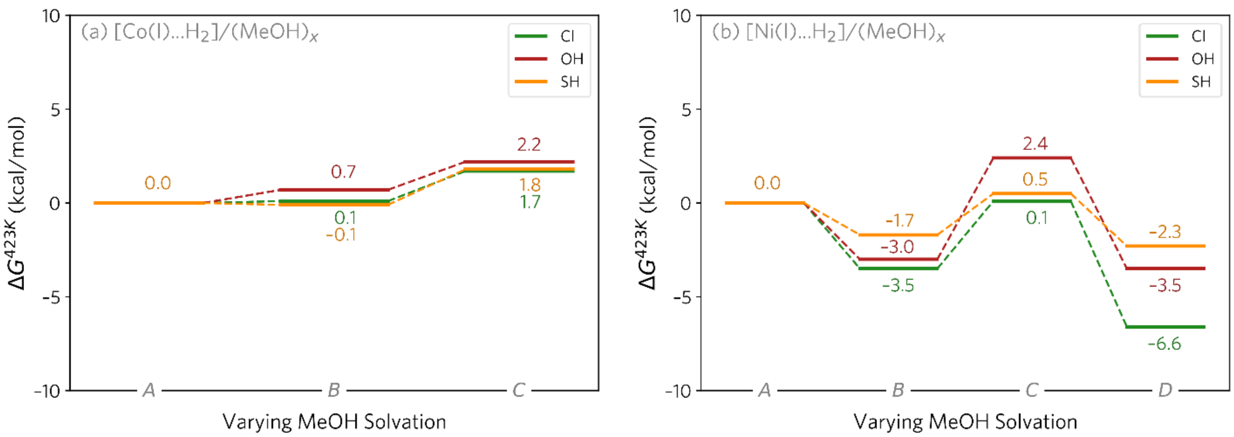


Supplementary Figure 46. Energetics of MeOH coordination at a $[M^{II}]^+$ species is shown for both Co (a) and Ni (b). Species A – F are demonstrated in Supplementary Figure 45. Overall, MeOH solvation at Ni is more favored compared to at Co for all coordination modes. At 423 K, the lowest energy structure for $[Co^{II}]^+$ is the solvated structure E, having three MeOH coordination at the open-metal site. For $[Ni^{II}]^+$, it is the solvated structure F that is analogous to E, having three MeOH coordination at the open-metal site, but also the other Ni center octahedrally coordinated by two more MeOH molecules. Gibbs free energies (at 150 °C) are given in kcal/mol, calculated at the M06-D3(0)/def2-TZVP/SMD(MeOH)//TPSSh-D3(BJ)/def2-SVP level of theory.

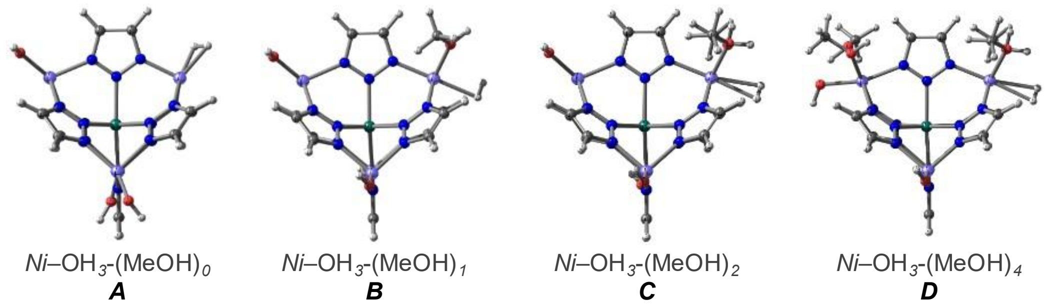
H.5. Effect of Systematic Methanol Solvation: H_2 Adsorption on 3D MOF Open Metal Sites $[M^I]$ and $[M^{II}]^+$



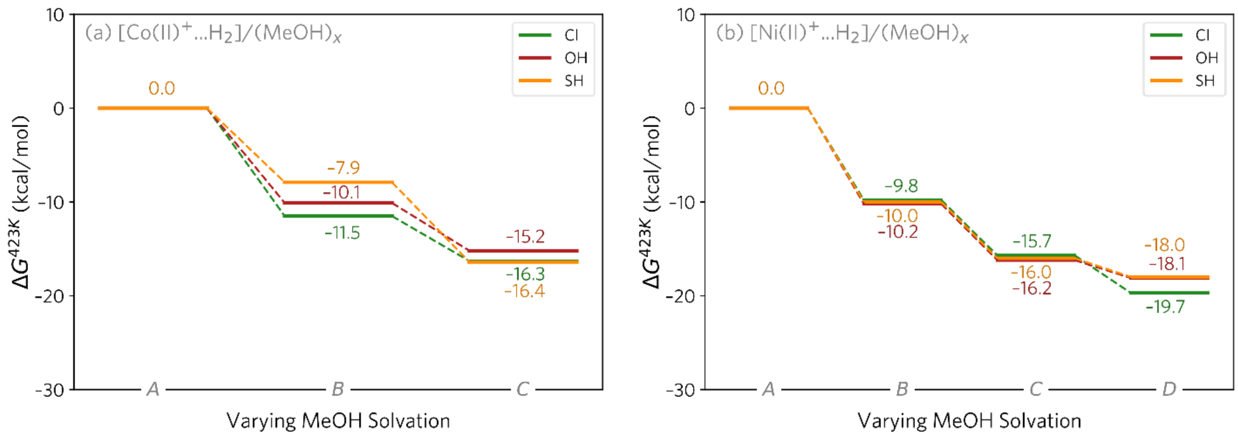
Supplementary Figure 47. Schematic representation of various coordination modes explored for MeOH binding to $[M^I..H_2]$ sites.



Supplementary Figure 48. Energetics of MeOH coordination for both Co (a) and Ni (b) at the $[M(I)..H_2]$ site are shown. For Co at 423 K, the unsolvated structure A (Supplementary Figure 47) is primarily relevant. In contrast, for Ni, species D (Supplementary Figure 47) is the most energetically favored state. Gibbs free energies (at 150 °C) are given in kcal/mol, calculated at the M06-D3(0)/def2-TZVP/SMD(MeOH)//TPSSh-D3(BJ)/def2-SVP level of theory.

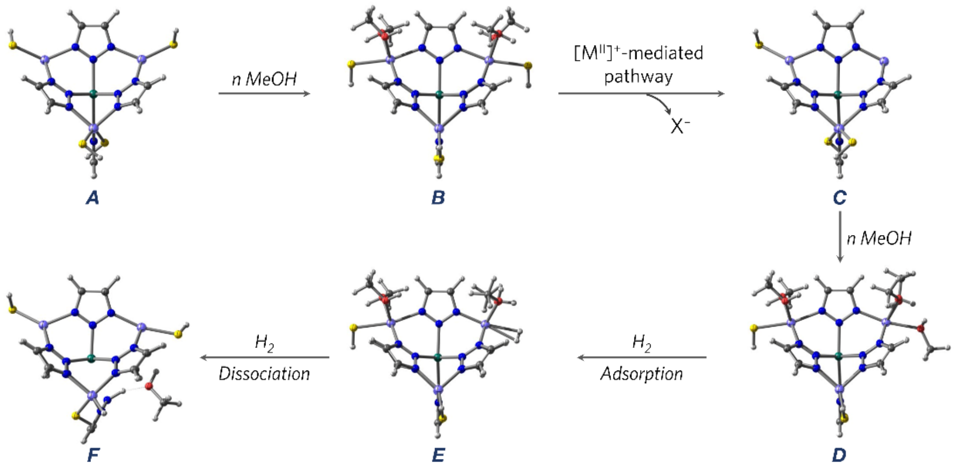


Supplementary Figure 49. Schematic representation of various coordination modes explored for MeOH binding to $[M^{II}..H_2]^+$ sites.

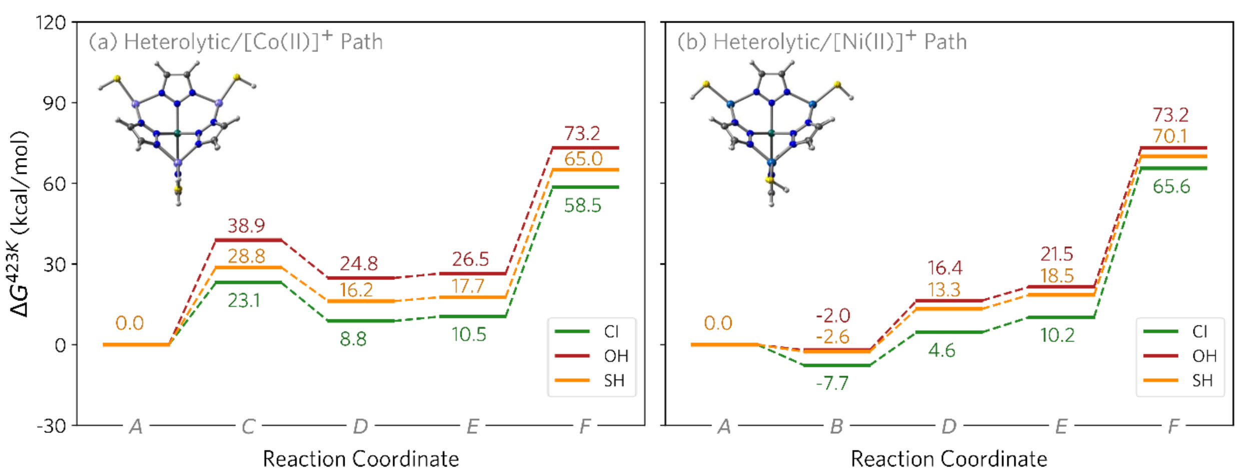


Supplementary Figure 50. Energetics of MeOH coordination for both Co (a) and Ni (b) at the $[M^{II}..H_2]^+$ site are shown. For Co at 423 K, structure C (Supplementary Figure 49) is primarily relevant. In contrast, for Ni, species D (Supplementary Figure 49) is the most energetically favored state. Gibbs free energies (at 150 °C) are given in kcal/mol, calculated at the M06-D3(0)/def2-TZVP/SMD(MeOH)//TPSSh-D3(BJ)/def2-SVP level of theory.

H.6. Summary of Heterolytic $[M^{II}]^+$–Mediated Path 3D MFU-4*l* catalyst



Supplementary Figure 51. Schematic for $[M^{II}]^+$–mediated open metal-site generation, hydrogen adsorption and dissociation for subsequent Ar–NO_2 reduction in the MFU-4*l* catalyst.



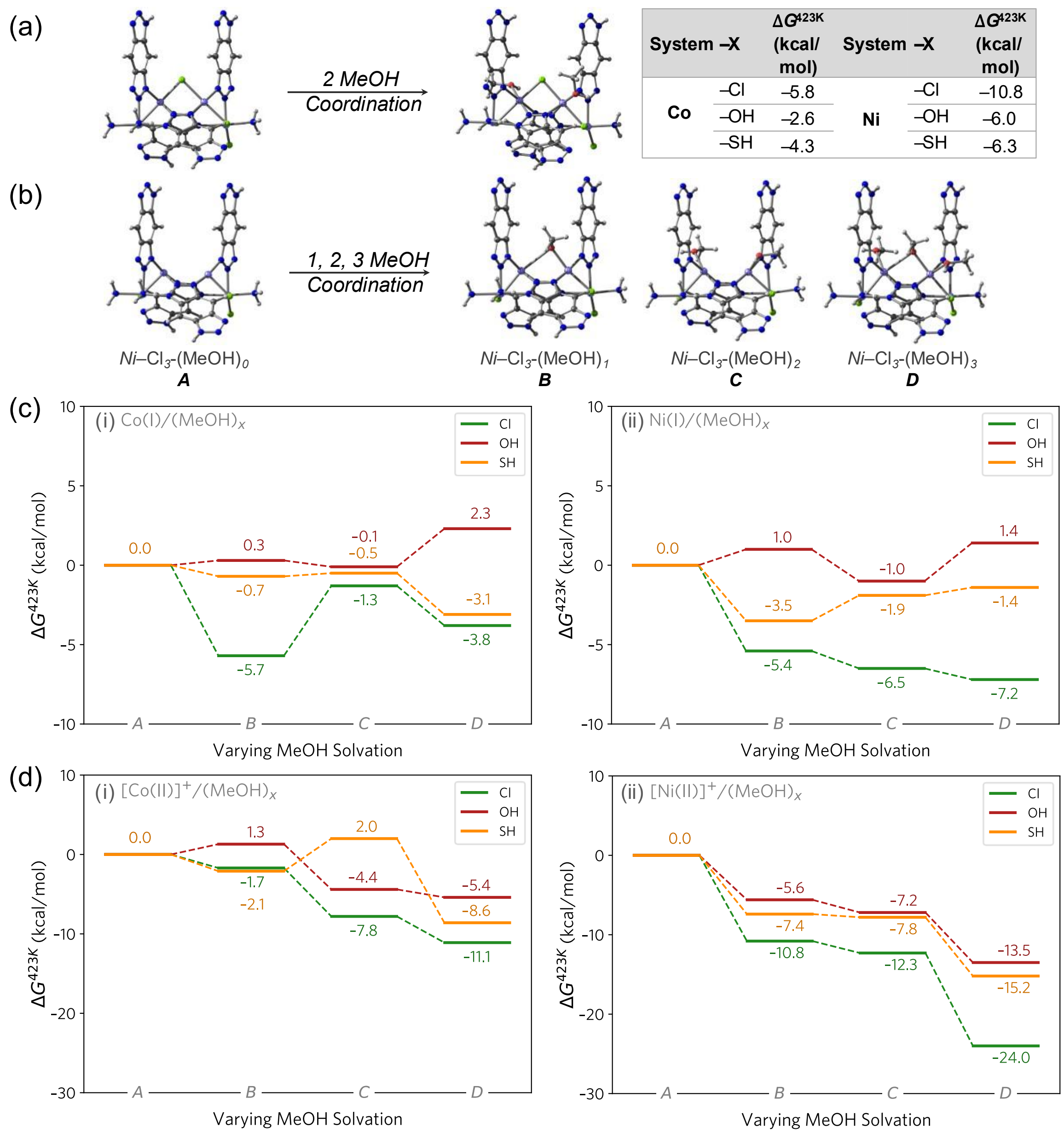
Supplementary Figure 52. $[M^{II}]^+$–mediated pathway for active hydrogen source (**F**) generation from M-MFU-4*l*-X catalysts (X = SH (yellow), OH (red), and Cl (green) lines). Gibbs free energies (at 150 °C) are given in kcal/mol, calculated at the M06-D3(0)/def2-TZVP/SMD(MeOH)//TPSSh-D3(BJ)/def2-SVP level of theory.

H.7. 1D BBTA System: Spin Ladder

Supplementary Table 17. Relative electronic energies (ΔE in kcal/mol) of the spin isomers for the 1D system with a four-metal center combination (not considering Zn), analogous to the one shown in Supplementary Figure 38. The energies are calculated at the TPSSh-D3(BJ)/def2-TZVP/SMD(MeOH)//TPSSh-D3(BJ)/def2-SVP level of theory.

System	–X	ΔE (kcal/mol)			
		Spin-A	***Spin-B***	***Spin-C***	***Spin-D***
Co	–Cl	0	0	13.5	48.9
	–OH	0	–1.1	–0.4	40.6
	–SH	0	–0.2	12.4	26.9
Ni	–Cl	0	0.1	0.1	55.4
	–OH	0	0	0	49.2
	–SH	0	0.4	0.3	41.2

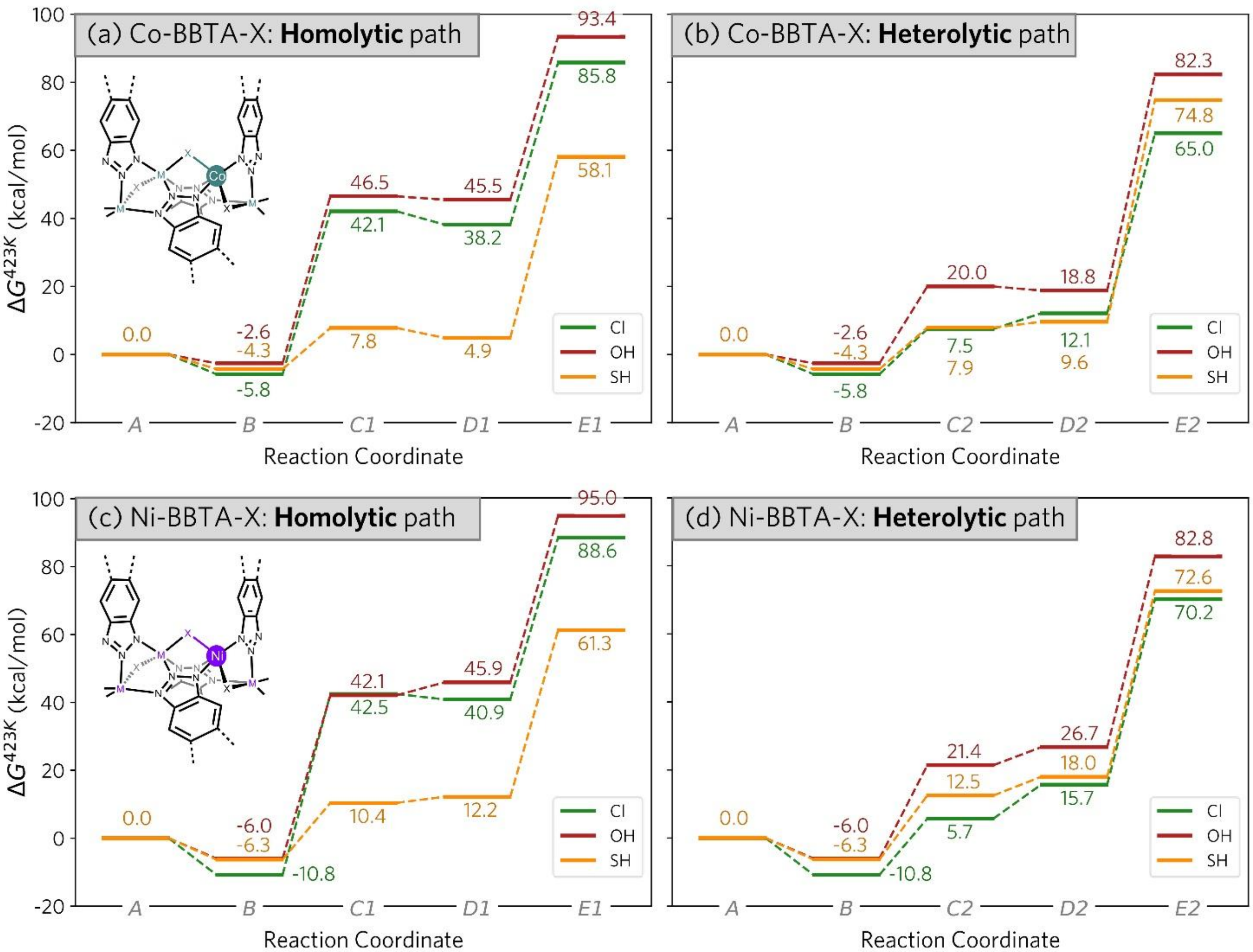
H.8. 1D System: Probing the Effect of Explicit MeOH Coordination



System	–X	ΔG^{423K} (kcal/mol)	System	–X	ΔG^{423K} (kcal/mol)
Co	–Cl	–5.8	Ni	–Cl	–10.8
	–OH	–2.6		–OH	–6.0
	–SH	–4.3		–SH	–6.3

Supplementary Figure 53. Various MeOH solvated structures explored for the 1D system, analogous to the 3D system. (a) Neutral Co(II) and Ni(II) species. For both metals, the table on the right shows that the coordination of 1 MeOH molecule is favored. (b) Corresponding MeOH coordination at a $[M^{I}]$ or $[M^{II}]^{+}$ species having an open metal site. Energetics for the species shown in panel *b*, both for the $[M^{I}]$ species (c) and for $[M^{II}]^{+}$ species (d) and both for Co (subpanel *i*) and Ni (subpanel *ii*). For $[M^{I}]$ species, a clear trend in MeOH solvated species could not be recognized. 2 MeOH molecule coordinated Species **C** was used as the resting state for both Co(I) and Ni(I) for consistency. For $[M^{II}]^{+}$ species, however, Species **D** was clearly found to be the most stable resting state for both metals. Gibbs free energies (at 150 °C) are given in kcal/mol, calculated at the M06-D3(0)/def2-TZVP/SMD(MeOH)//TPSSh-D3(BJ)/def2-SVP level of theory.

H.9. Summary of Homolytic $[M^{I}]$–and Heterolytic $[M^{II}]^{+}$–Mediated Path 1D Ni-BBTA Catalyst



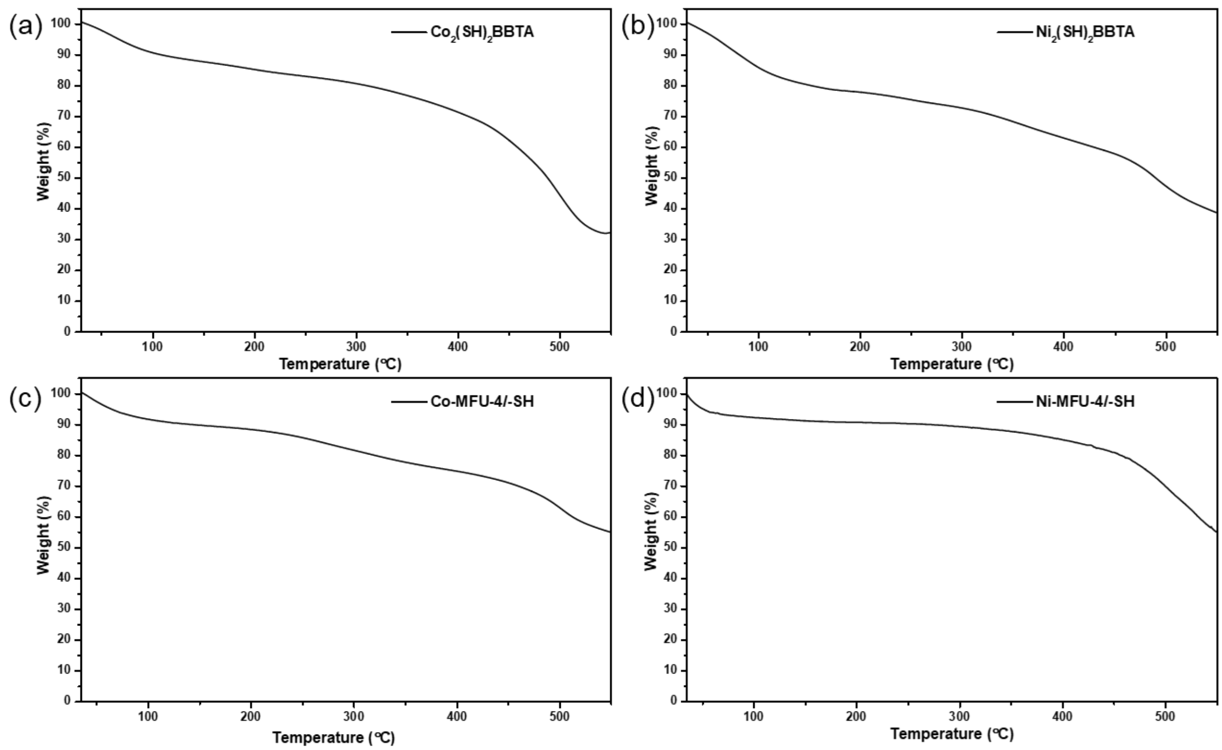
Supplementary Figure 54. Reaction profiles for active hydrogen source generation (**E1** and **E2**) in the reduction catalysis with M-BBTA-X catalysts. (a-b) M = Co; (c-d) M = Ni. The calculated energetics indicate a cation-mediated pathway for X = Cl (green) and OH (red), while a radical pathway involving a M(I) is predicted for X = SH (yellow). Gibbs free energies (at 150 °C) are calculated at the M06-D3(0)/basis-II/SMD(MeOH)//TPSSh-D3(BJ)/basis-I level of theory.

H.10. Energy decomposition analysis (EDA)

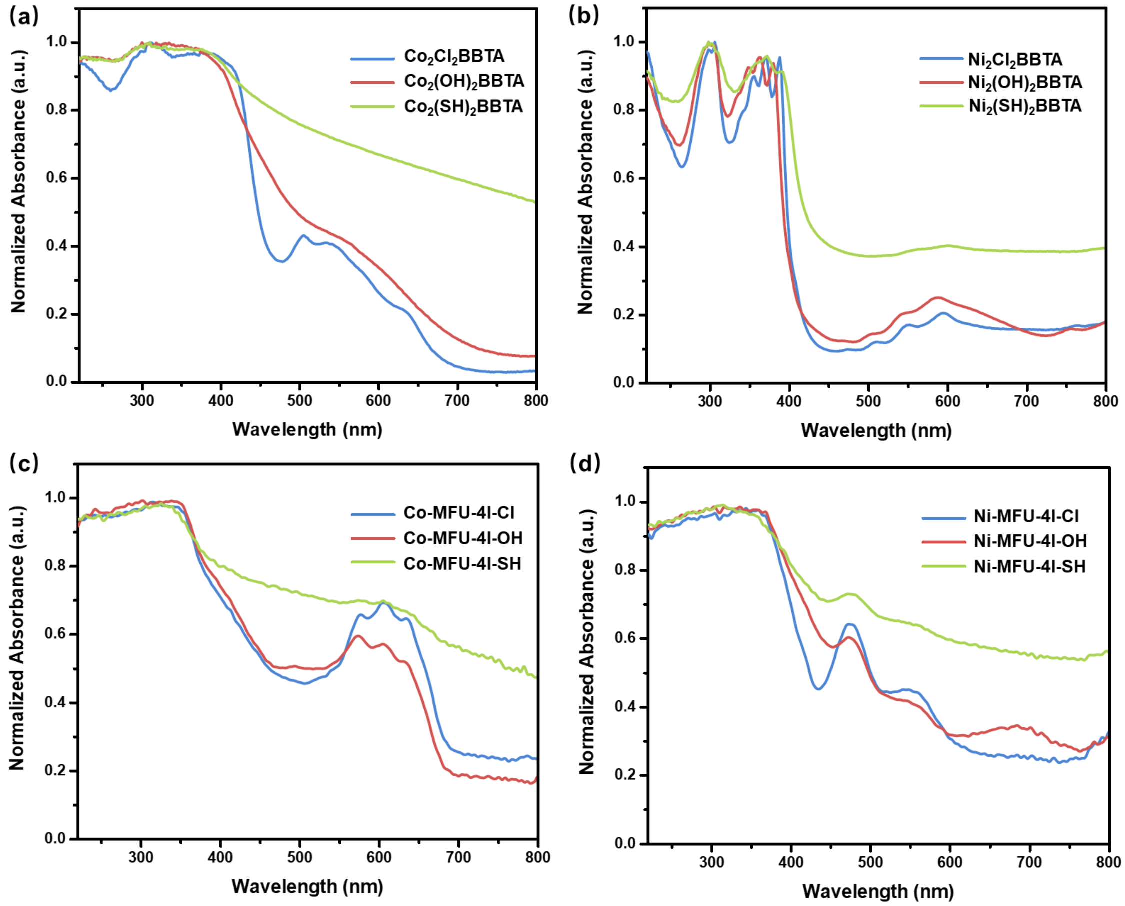
In the ALMO-EDA framework (ALMO = absolutely localized molecular orbitals; EDA = energy decomposition analysis),[4,5] the total electronic contribution (ΔE_{Tot}) for binding X(•) to a formal M(I) site, forming an M(II)–X site, comprises three fundamental components (Extended data Table 1).[6] The frozen term (ΔE_{Frz}) represents the energy change when bringing two fragments together without altering their electron distributions. The polarization term (ΔE_{Pol}) accounts for the energy reduction when the electron distribution of each fragment adjusts to the other's presence, without electron sharing. Lastly, the charge transfer term (ΔE_{CT}) quantifies the additional energy lowering from orbital mixing between the two fragments.

To probe the differential reactivity of Co^{II}–SH and Ni^{II}–SH active sites in BBTA and MFU-4*l* frameworks we specifically probed the "frozen" energy (ΔE_{Frz}) between the catalyst (species **C**) and approaching H_2, decomposing this interaction energy term into three components (Extended data Table 2):[7,8] the electrostatic term ΔE_{Elec} for interactions between fragment charge distributions, ΔE_{Pauli} accounting for repulsion between overlapping filled orbitals, and the dispersion interaction term ΔE_{Disp}.

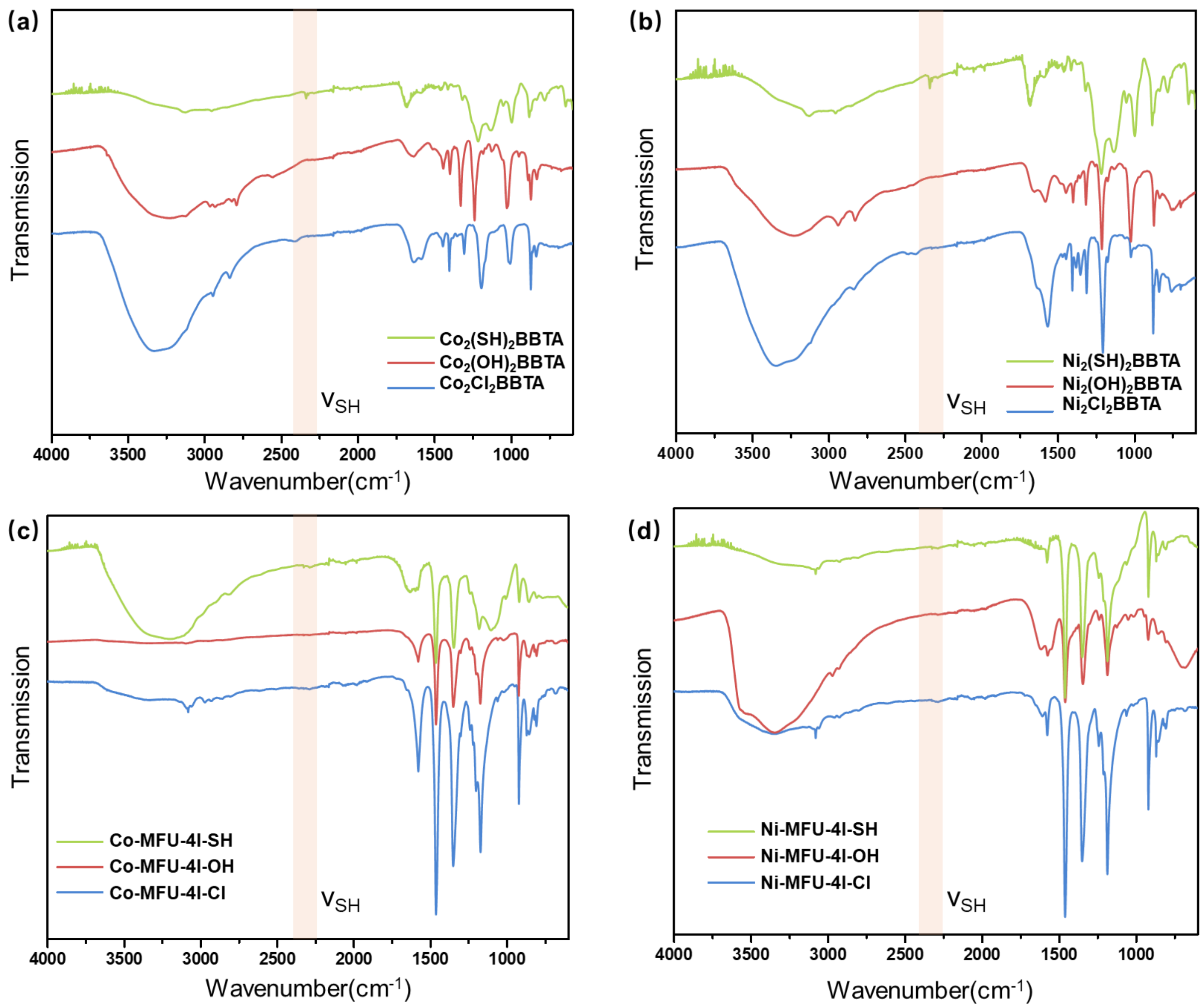
Section I. Infrared Spectra, UV-vis Spectra, TGA and Elemental Analysis Results with ICP-OES



Supplementary Figure 55. Thermogravimetric analysis (TGA) of (a) $Co_2(SH)_2BBTA$, (b) $Ni_2(SH)_2BBTA$, (c) Co-MFU-4*l*-SH, and (d) Ni-MFU-4*l*-SH. The decrease in mass below 100 °C is due to the desorption of guest molecules from the MOF cavity.



Supplementary Figure 56. Solid-state UV-vis Spectra of (a) Co_2X_2BBTA, (b) Ni_2X_2BBTA, (c) Co-MFU-4*l*-X and (d) Ni-MFU-4*l*-X (X= Cl, OH, SH)



Supplementary Figure 57. Infrared Spectra of (a) Co_2X_2BBTA, (b) Ni_2X_2BBTA, (c) Co-MFU-4*l*-X and (d) Ni-MFU-4*l*-X (X= Cl, OH, SH)

Supplementary Table 18. Elemental analysis of bulk samples with ICP-OES

MOF			
$Co_2(SH)_2BBTA$	$C_6N_6H_2Co_2S_2H_2$		
Element	Theoretical Value	Experimental Value	Molar Ratio
Co	34.40%	29.6%±0.5%	1.0
S	18.70%	16.7%±0.4%	1.0
$Ni_2(SH)_2BBTA$	$C_6N_6H_2Ni_2S_2H_2$		
Element	Theoretical Value	Experimental Value	Molar Ratio
Ni	34.40%	29.7%±1.2%	1.0
S	18.80%	18.6%±1.7%	1.1
Co-MFU-4*l*-SH	$ZnCo_4(C_{12}O_2N_6H_4)_3(SH)_4(H_2O)_8$		
Element	Theoretical Value	Experimental Value	Molar Ratio
Co	17.40%	16.0%±0.2%	3.9
Zn	4.80%	4.8%±0.1%	1.1
S	9.40%	9.1%±0.3%	4.1
Ni-MFU-4*l*-SH	$Zn_{2.5}Co_4(C_{12}O_2N_6H_4)_3(SH)_4(H_2O)_8$		
Element	Theoretical Value	Experimental Value	Molar Ratio
Ni	10.7%	11.1%±1.0%	2.6
Zn	11.9%	11.8%±0.8%	2.4
S	9.4%	9.0%±0.3%	3.7
$Co_2(SH)_2BBTA$-2.0eq	$C_6N_6H_2Co_2S_2H_2$		
Element	Theoretical Value	Experimental Value	Molar Ratio
Co	28.9%	28.0%±3%	1.0
S	31.4%	28.6%±1%	1.9

Section J. References

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